GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation

Yuhang Jiang, Guohui Deng, Miaozhong Xu, Chao Ruan, Jinling Zhao, *Member, IEEE*, and Linsheng Huang

***Abstract*—Referring remote sensing image segmentation (RRSIS) segments the object named by a natural-language expression in an aerial image. Existing training-free methods resolve the expression through implicit vision–language activations or region–text similarity, which gives weak control over the spatial, superlative, and ordinal relations that dominate aerial referring, such as *the rightmost ship* or *the second court from the left*. We propose GeoSelect, a training-free pipeline that reframes referring as the execution of a typed spatial program. A frozen, text-only language model synthesizes the expression into a small domain-specific language (DSL), a well-formedness checker accepts the program, and a deterministic executor runs it. The central abstraction is a single *scored candidate set* type under which every operator composes: continuous geometric fields realize position and proximity, while discrete set and order operators add the extremum, ordinal, top-k, and relational constructions that fields alone cannot express. Execution is explicit, so every intermediate is inspectable, and a reliability ladder degrades any failing program to the field-only special case. GeoSelect achieves 58.86 mIoU on RRSIS-D test and 55.27 mIoU on RISBench test, more than twice the best prior training-free method on RRSIS-D, with no referring supervision and on a single GPU. Under a fixed detector and segmenter, explicit execution improves over implicit selectors under the same backbone; the best-box-IoU and outcome-partition diagnostics motivate complementary tests of proposal recall and program-path behavior, with the program path the clearer priority on RISBench, and an exposure audit shows comparable accuracy on the audited unseen subset. Code and configurations are available at https://github.com/Avalon-S/GeoSelect**

***Index Terms*—Neuro-symbolic reasoning, program synthesis, referring image segmentation (RIS), remote sensing, spatial reasoning, training-free, vision–language models.**

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I. Introduction

REFERRING remote sensing image segmentation (RRSIS) [1] requires instance-level disambiguation in crowded aerial scenes. Expressions frequently specify the referent through spatial, superlative, and ordinal cues such as *the upper right ship*, *the rightmost tank*, or *the second court from the left*, where several same-class objects would otherwise be valid candidates. Resolving the referent therefore demands that the relational content of the expression be evaluated, not merely that the object category be recognized.

Recent supervised networks [1], [2], [3] reach strong accuracy on RRSIS-D, but require dense referring masks and a separate training run for each benchmark. Training-free methods instead combine frozen pretrained components, yet the best published result on RRSIS-D test remains far below the supervised specialists. The gap suggests that pretrained perception alone does not solve spatial referring.

We study the representation of the spatial constraint (Fig. 1). Existing training-free pipelines derive the referent from vision–language activations, attention, or region–text similarity. These signals may identify the relevant object category, but do not directly execute relations such as *left of*, *near*, *rightmost*, or *second from the left*. In a controlled selector comparison with detection and segmentation held fixed, explicit geometric execution outperforms the stronger of two implicit selectors, and the gap is concentrated on the spatial and compositional expressions and nearly disappears on the pure-attribute subset. This advantage widens sharply with same-class clutter; overhead imagery, which packs many near-identical instances into a single scene, makes this clutter regime the rule rather than the exception. Interinstance geometric selection is thus where aerial referring is hardest, and where explicit execution helps most, while recognition-only signals are weakest.

Our starting point is a field-only special case that represents each expression as a fixed relation over a closed lexicon of positional and proximity predicates, composes their pixel-level fields, and ranks candidate boxes by the composed prior. This covers image-relative, object-anchored, and attribute expressions, but it cannot express an extremum, a rank, or a counted subcollection: these operate on the candidate set rather than on fields over the image plane, and they dominate the longer

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Fig. 1. Why explicit execution. For a spatial expression over several same-class objects, implicit region–text matching (left, GeoRSCLIP) selects the wrong instance, whereas GeoSelect synthesizes and executes a typed spatial program that scores the candidate set (middle, brighter = higher score; the chosen box in green) and returns the instance matching the ground truth (right). Same candidate boxes and segmenter; only the selector differs. Yellow outline: ground-truth target.

expressions of RISBench. GeoSelect removes this ceiling by reframing referring as explicit spatial-program execution. A frozen, text-only language model synthesizes a typed program in a small domain-specific language (DSL); a well-formedness checker accepts or rejects it; and a deterministic executor runs it over a single scored-candidate-set type that unifies the continuous field algebra with discrete set and order operators. Position and proximity remain dense geometric fields, while superlative, ordinal, and compositional expressions are synthesized from extremum, ordinal, top-k, and relation operators that the closed lexicon lacked. Every intermediate program, field, and ranking is inspectable, and an explicit reliability ladder degrades any ill-formed or failing program to the field-only selector, so a malformed program forfeits no answer. Although executing a program over perception modules has precedent in natural-image referring [4], [5], GeoSelect is built for the aerial regime rather than ported from it. Overhead scenes pack numerous same-class instances and complex geospatial relations that make instance disambiguation, not recognition, the core difficulty [1], [2], so the bottleneck is interinstance geometric selection; and open-vocabulary detectors pretrained on natural scenes transfer poorly to overhead imagery [6], so candidates must come from an aerial-pretrained detector.

Contributions:

1) *(C1) Referring as Explicit Spatial-Program Execution:* We recast training-free RRSIS as executing a typed spatial program over a single scored-candidate-set type. With no referring supervision and on a single GPU, the pipeline reaches 58.86 mIoU on the RRSIS-D test, more than twice the best prior training-free result and about 93% of the supervised specialist RMSIN [1].
2) *(C2) Discrete Operators and Controlled Attribution:* Extremum, ordinal, top-k, and binary-relation operators make the superlative, ordinal, and compositional constructions that a closed positional lexicon cannot represent expressible in the first place. With the detector and segmenter fixed, explicit execution outperforms the implicit selectors, and the best-box-IoU proxy and the outcome partition suggest proposal recall and the program path as complementary directions for further gains.
3) *(C3) Cross-Domain Accuracy and a Leakage Audit:* The single frozen configuration transfers across domains to RISBench at 55.27 mIoU, more than 1.7 times the best prior training-free method. An image-level pretraining-exposure audit further shows that the exposure-audited unseen subset reaches 58.4 mIoU (54.5 under a conservative seen-set), still well above prior training-free methods; this exposure is quantified, not left unexamined.

Design choices are made on RRSIS-D val; the resulting configuration is frozen before any test evaluation and applied to RISBench with no test-set recalibration. Sections III–V present the formulation, method, and experiments.

II. Related Work

A. Referring Remote Sensing Image Segmentation

Referring image segmentation (RIS) originated in natural images, where CLIP-driven CRIS [7], the transformer LAVT [8], and earlier recurrent models [9], [10], [11] fuse a language embedding with dense visual features and decode a mask. The aerial setting introduces overhead viewpoints, large-scale variation, and dense same-class clutter, which motivated dedicated benchmarks and architectures. RMSIN [1] introduced the RRSIS-D benchmark and a rotation-aware multiscale interaction network; CroBIM [2] contributed RISBench and a cross-modal bidirectional interaction module; FIANet [3], LGCE [12], and BTDNet [13] strengthen cross-modal alignment and boundary quality; and RSRefSeg2 [14] adapts vision foundation models with SAM-based decoding. These supervised specialists reach roughly 61–69 mIoU on RRSIS-D and serve as the learned-model reference in our experiments, but each requires dense referring masks and a separate training run per benchmark, and most release distinct weights for RRSIS-D and RISBench. Our setting removes

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both requirements and instead composes frozen perception modules, trading learned end-to-end fusion for an explicit selection rule.

B. Training-Free Referring Grounding and Segmentation

Training-free RRSIS resolves the expression with frozen pretrained components. EKP-HRM [15] decomposes expressions hierarchically and scores FreeSOLO masks by GeoRSCLIP cosine similarity; RSVG-ZeroOV [16] couples Qwen2.5-VL with a diffusion grounder; DGL-RSIS [17] reads masks from enhanced Grad-CAM activations; and text-promptable segmenters such as SAM3 [18] can be queried with the expression directly. What these methods share is that the referring decision is taken implicitly, through vision–language activations, attention, or region–text similarity, rather than by explicitly evaluating the parsed spatial relation. The published training-free ceiling on the RRSIS-D test has remained far below the supervised specialists, which suggests that pretrained semantic matching alone does not resolve the spatial and ordinal content of aerial referring expressions. We adopt the same frozen-component constraint but make the relation an executable object. Adjacent training-free remote-sensing segmenters target other tasks: SegEarth-OV [19] and its SAM3-based successor SegEarth-OV3 [20] perform open-vocabulary semantic segmentation, and InstructSAM [21] pairs a frozen LVLM with SAM and a remote-sensing CLIP through structured output for instruction-oriented recognition; neither evaluates the relational referring decision we study. We use the term training-free in the strict sense of Section III-B: trained RRSIS vision–language systems [22], [23], [24] are excluded even when evaluated zero-shot, and a concurrent SAM-prompting method [25] achieves its strongest RRSIS-D result (67.6 mIoU) only by LoRA-tuning its language model; its fully training-free variant reaches 24.9, below GeoSelect's 58.86.

C. Neuro-Symbolic Reasoning and Visual Program Synthesis

Resolving a compositional query by executing a program over perception modules has a long lineage in natural images. Neural module networks [26] assemble a layout of learned modules per question; NS-VQA [27] and the neuro-symbolic concept learner [28] run a symbolic program over an object-centric scene representation; and prompting-era methods such as VisProg [4] and ViperGPT [5] have a frozen LLM emit a program over off-the-shelf vision APIs. For referring specifically, ReCLIP [29] decomposes an expression into subject, relation, and object and scores candidate regions with frozen CLIP. GeoSelect shares this family's spirit of a frozen LLM that synthesizes a typed, deterministically executed program, but differs in the *executed representation*. VisProg and ViperGPT dispatch to neural black-box subroutines whose spatial reasoning is opaque and whose outputs carry no type or correctness guarantee, and ReCLIP scores relations through CLIP similarity rather than geometry. In GeoSelect, each spatial predicate is a closed-form geometric field or set/order operator over a single scored-candidate-set type (2), so every intermediate field and ranking is inspectable and checked rather than dispatched to an opaque subroutine. The result is a neuro-symbolic referring pipeline whose symbolic half is an explicit geometric calculus rather than a set of learned modules.

D. Spatial Relation Modeling, Detection, and Promptable Segmentation

The geometric fields we execute connect to a body of work on representing spatial language as graded, composable maps. Fuzzy spatial-relation models render directional and distance terms as membership fields and combine them with fuzzy connectives [30], [31], and cognitive spatial templates model where a relation places its referent relative to an anchor [32]. Our monotone and extremal kernels are a closed-form, benchmark-calibrated instance of this idea, connected to a text-only program synthesizer and an aerial detector. In remote sensing, scene-graph methods [33], [34], [35] predict dense relation graphs over a whole image, whereas RRSIS grounds one queried relation; RSGround-R1 [36] and ProVG [37] decompose referring expressions into reasoning steps but train their grounding models. Candidate generation relies on open-vocabulary detection: generic detectors such as GLIP [38] and Grounding-DINO [39] transfer poorly to overhead imagery, while LAE-DINO [6] pretrained on the aerial LAE-1M corpus achieves substantially stronger open-set detection on DIOR, and remote-sensing vision–language encoders [40], [41], [42] provide semantic matching but not precise spatial execution. The final mask comes from a promptable segmenter [18], [43], [44], [45]; GeoSelect treats this stage as interchangeable and assigns the referring decision entirely to the executed program.

Remote sensing foundation models and in-context segmenters address related prompting settings. SkySense++ [46] learns transferable multimodal Earth-observation representations, while SkySense-VITA [47] supports cross-image visual, textual, and fused prompting. GeoSelect instead maps relational referring expressions to an executable spatial program.

III. Problem Formulation and Taxonomy

This section fixes the task, the training-free setting, the selection-by-candidates formulation that our method instantiates, and a taxonomy of referring expressions that organizes the experimental analysis. The taxonomy is defined on the language alone and is independent of any method, so it partitions a fixed benchmark identically for every system we compare.

A. Task Definition

Given an aerial image $I \in \mathbb{R}^{H\times W\times 3}$ and a natural-language referring expression E, RRSIS produces a binary mask $M \in \{0,1\}^{H\times W}$ of the single object that E designates. The expression typically names an object category together with disambiguating cues: an *attribute* (*"the gray harbor"*) or a *position* relative to the image frame or to another object (*"the ship in the upper right"*, *"the vehicle left of the building"*). A third cue is a *set restriction* that singles out one instance from several of the same class (*"the rightmost tank,"* *"the second court from the left"*). The defining difficulty of RRSIS, relative

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to category segmentation, is that several same-class objects are usually present and only the relational and ordinal content of E separates the referent from its distractors.

B. Training-Free Setting

We call a method *training-free* when no parameter of any component is updated on referring-segmentation supervision from the target benchmark; pretrained components may retain their original pretraining. Formally, a system is a composition of modules $f_{\theta_1}, \ldots, f_{\theta_m}$ whose weights θ_i are frozen at their public checkpoints, with no gradient taken on RRSIS-D or RISBench image–expression–mask triples. Under this definition, a vision–language model fine-tuned for referring segmentation is not training-free even when evaluated zero-shot on a held-out split, whereas an open-vocabulary detector that retains its detection pretraining is admissible because pretraining is not referring supervision. This setting removes per-benchmark mask annotation and per-benchmark training runs, at the cost of having to define the referring decision explicitly rather than learning it.

C. Selection by Candidates

We factor RRSIS into three training-free stages and locate the referring decision entirely in the middle one

$$\underbrace{\mathcal{B} = \mathcal{D}(I,\, Q(E))}_{\text{candidate generation}} \longrightarrow \underbrace{b^\star = \Pi(E, \mathcal{B}, I)}_{\text{selection}} \longrightarrow \underbrace{M = \mathcal{S}\left(I, b^\star\right)}_{\text{segmentation}}. \tag{1}$$

A frozen open-vocabulary detector $\mathcal{D}$, queried with category and attribute terms $Q(E)$ extracted from the expression, returns a set of candidate boxes $\mathcal{B} = \{b_1, \ldots, b_N\}$; a selection operator Π chooses the referent box $b^\star$; and a frozen promptable segmenter $\mathcal{S}$ turns $b^\star$ into the mask. Stages $\mathcal{D}$ and $\mathcal{S}$ are standard frozen perception modules; the scientific content of a training-free method is the design of Π, which must evaluate the spatial, superlative, and ordinal content of E without learned parameters. Implicit methods realize Π as a region–text similarity (rank b_j by the cosine between a vision–language embedding of the crop b_j and of E) or as a detector-confidence ranking; both reduce the relation in E to a single matching score. We instead realize Π as an executable program, described in Section IV.

D. Scored Candidate Sets

To unify continuous geometric scoring with discrete set and order operations, we type every intermediate value as a *scored candidate set*

$$\mathcal{C} = \left\{\left(b_j,\, s_j\right)\right\}_{j=1}^{N}, \qquad b_j \in \mathbb{R}^4,\ s_j \in [0, \infty) \tag{2}$$

a list of boxes each carrying a non-negative score. The detector output is the initial set with s_j the detection confidence; a continuous geometric predicate reweights the scores in place; a discrete operator such as ARGMAX or NTH reranks or prunes the set; and conjunction and disjunction combine two sets by score algebra. Every operator returns a scored candidate set, so continuous fields and discrete combinators compose freely in any order, and the final referent is the top-scoring element $b^\star = \arg\max_j s_j$. This single type is what lets one executor host both the field algebra inherited from continuous geometry and the set/order operators that the longer expressions require.

E. Expression Taxonomy

We group expressions by the reasoning the selector must perform, using a deterministic, method-independent text rule applied to E alone. Three categories follow prior RRSIS practice and three extend it to the compositional constructions that motivate this work.

1) *Image-Relative:* Position relative to the image frame (*"the field in the lower left"*). The anchor is always available.
2) *Object-Anchored:* Position relative to another named object (*"the car to the right of the roundabout"*), requiring the anchor to be detected and bound to the correct instance.
3) *Attribute:* Disambiguation by an intrinsic property with no spatial relation (*"the white storage tank"*); the selector reduces to the detector ordering.
4) *Superlative:* An extremum over a set (*"the rightmost ship," "the top-most court"*), requiring an ARGMAX/ARGMIN over an axis.
5) *Ordinal:* A ranked position within a set (*"the second vehicle from the left"*), requiring an NTH selection.
6) *Compositional:* Two or more of the above chained (*"the building in the upper left near the river"*), requiring nested execution of fields and operators.

The first three are base types expressible by continuous fields alone; the last three are operator constructions that a closed positional lexicon cannot represent and that the discrete operators of Section IV-D target. Each expression carries exactly one base tag together with any operator tags it triggers, assigned by a deterministic text rule. On a stratified sample of 300 validation expressions labeled independently by two annotators, the two agree perfectly on the base frame type and on the ordinal, superlative, and compositional tags (Cohen's $\kappa = 1.0$), and substantially on the multi-instance tag ($\kappa = 0.66$), which is the borderline case in these single-instance benchmarks. The text rule reproduces these human labels on 98.0% of ordinal, 96.3% of superlative, 79.0% of compositional, and 87.0% (annotator A) and 86.0% (annotator B) of multi-instance tags. The base frame type is a three-way label (image-relative, object-anchored, attribute) that the text rule reproduces on 67.3% of the sample; separating a frame-relative cue from an object-anchored one from text alone is the hard case, so we treat this stratification as noisy. These tags are used only for descriptive stratification and never for prediction. The operator strata therefore overlap the base strata and one another, so per-stratum counts need not sum to the split size. The tags are derived from text and never from predictions, so stratifying any method's per-sample scores by tag rebuckets fixed numbers and leaves every overall metric unchanged.

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F. Evaluation Protocol

We follow the RMSIN protocol [1]: predicted and ground-truth masks are resized to 480 × 480 by nearest-neighbor interpolation before scoring, so that all methods are compared at a common resolution. We report mean intersection-over-union (mIoU, the per-sample IoU averaged over the split, equivalent to gIoU), overall IoU (oIoU, the ratio of summed intersection to summed union, equivalent to cIoU), and precision Pr@X at IoU thresholds $X \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. mIoU weights every expression equally and is our primary metric; oIoU is dominated by large referents and exposes under-coverage of big objects by the single-mask decoder.

IV. Method

A. Overview

GeoSelect resolves a referring expression E as the execution of a typed spatial program (Fig. 2; the full procedure is Algorithm 1). A frozen, text-only language model synthesizes a program π in a small DSL from the expression alone (Section IV-B). A well-formedness checker accepts or rejects π. A legal program is run by a deterministic executor that evaluates each operator over the scored candidate sets of (2), so continuous geometric fields (Section IV-C) and discrete set and order operators (Section IV-D) run under one dispatch loop. If a program is ill-formed, fails, or returns nothing, a reliability ladder (Section IV-H) falls back to the continuous-field selector, recovering the field-only result exactly rather than losing the answer. The top-scoring candidate is segmented by a frozen promptable decoder. No parameter of the parser, detector, or segmenter is updated on referring supervision. Concretely, GeoSelect instantiates the abstract selection stage Π of (1) as the synthesize-then-execute factorisation

$$\begin{aligned} \pi &= \text{Synthesize}(E) \\ b^\star &= \text{Execute}(\pi, \text{Detect}(I)) \\ M &= \text{Segment}(I, b^\star) \end{aligned} \tag{3}$$

where only SYNTHESIZE reads language and only DETECT and SEGMENT read pixels, so the spatial reasoning is isolated in the symbolic executor.

Three design principles organize the method.

1) *One Closed Type:* Every operator maps a scored candidate set to another (2), so continuous fields and discrete operators compose and nest under a single type instead of separate stages.

2) *Closed-Form Geometry:* Each spatial predicate is an explicit geometric computation, not a learned or similarity-scored module, so every intermediate field and ranking is inspectable.

3) *Reliability Floor:* A field-only special case of the same algebra is retained as a deliberate component that any failing program falls back to, so expressive power never costs an answer.

Principle 3) is concrete. Our field-only special case maps each expression to a fixed relation over positional and proximity predicates and composes their fields by min / max. That representation covers image-relative, object-anchored, and attribute expressions but not the extremum, rank, and top-k constructions, which act on the candidate set rather

Algorithm 1 GeoSelect: Synthesize, Check, Execute, Fall Back

Require: expression E, image I; frozen LLM, detector DETECT, segmenter SEGMENT
Ensure: referent mask M

$\pi \leftarrow$ SYNTHESIZE (E) *text-only LLM; the image is unseen*
$\pi \leftarrow$ REPAIR (π) *fix benign synthesis slips, model-free*
if WELLFORMED(π) **then**
  $\mathcal{C} \leftarrow$ EXEC(π) *run the typed program*
  if $\mathcal{C} = \varnothing$ **or** execution errs **then**
    $\mathcal{C} \leftarrow$ FIELDONLY (E, I) *fall back: execution failed*
  end if
else
  $\mathcal{C} \leftarrow$ FIELDONLY (E, I) *fall back: ill-formed program (Sec. IV-H)*
end if
$b^\star \leftarrow \arg\max_{b_j \in \mathcal{C}} s_j$
return $M \leftarrow$ SEGMENT $(I, b^\star)$
function EXEC(node)
  if node is SELECT (t) **then**
    return DETECT (I, t) *initial candidate set* $\mathcal{C}$
  else if node is AND/OR/NOT **then**
    return min / max /comp of children *field algebra, Eq. (7)*
  end if
  $\mathcal{C} \leftarrow$ EXEC(node.child)
  if node is FILTER (ϕ, A) **then**
    $\mathbf{A} \leftarrow$ RESOLVEANCHOR(A); $P \leftarrow$ FIELD $(\phi, \mathbf{A})$
    $s_j \leftarrow s_j P(\text{center}\, b_j) \;\; \forall b_j$ *Eq. (6)*
  else if node is ARGMAX/ARGMIN (u) **then**
    extremal-softmax rescore on axis u *superlative, Eq. (8)*
  else if node is NTH (n, u) **then**
    rank on u, keep the n-th *ordinal, Eq. (9)*
  else if node is RESTRICTCOUNT (k) **then**
    keep top-k candidates *top-k candidate restriction, Eq. (10)*
  else if node is RELATE (r, A) **then**
    $\mathbf{A} \leftarrow$ RESOLVEANCHOR(A); $s_j \leftarrow s_j \max_{a \in \mathbf{A}} r(b_j, a)$ *relation, Eq. (11)*
  end if
  return $\mathcal{C}$
end function
function RESOLVEANCHOR(a)
  if a = IMAGE **then**
    return image frame
  else if a is a bare SELECT **then**
    return all detected instances *marginalize*
  else
    return top-scoring box of EXEC(a) *constrained anchor*
  end if
end function

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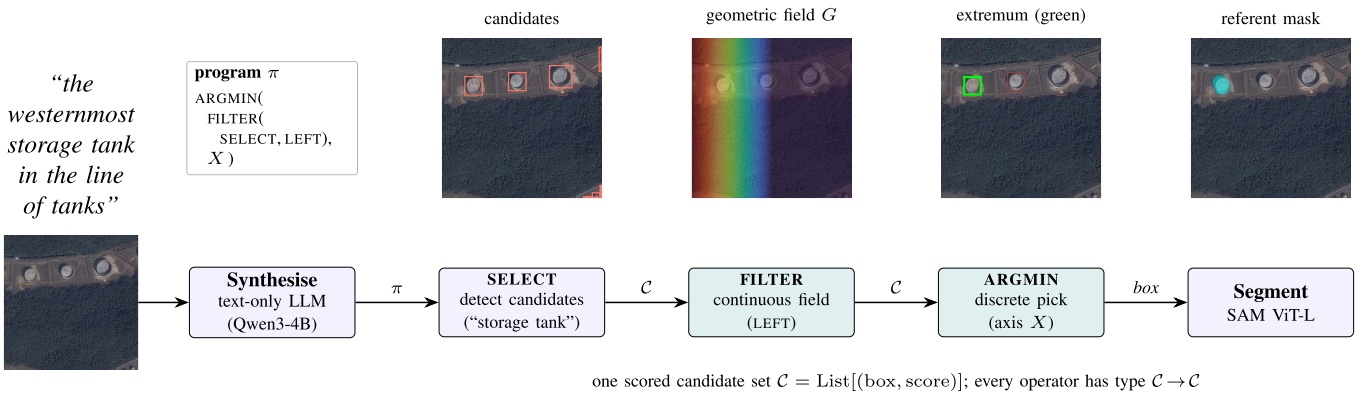


Fig. 2. GeoSelect executes a typed spatial program over a single scored candidate set, with a *real* per-operator trace above each stage (RISBench, "the westernmost storage tank in the line of tanks"). A frozen text-only LLM synthesizes the program π; the executor evaluates it bottom-up: SELECT returns the detected candidates, the continuous FILTER (teal) reweights them by a geometric field, and the discrete ARGMIN (teal) rescores the set to its extremum (green), which SAM segments.

```
prog   ::= expr
expr   ::= Select(phrase)
         | Filter(expr, pred, anchor)
         | Argmax(expr, axis) | Argmin(expr, axis)
         | Nth(expr, n, axis)
         | RestrictCount(expr, k)
         | Relate(expr, rel, anchor [, anchor2])
         | And(expr, expr) | Or(expr, expr)
         | Not(expr)
anchor ::= IMAGE | expr
pred   ::= LEFT|RIGHT|TOP|BOTTOM|UL|UR|LL|LR
           | CENTER|NEAR
rel    ::= NEAR|BETWEEN|ADJACENT|CONTAINS|INSIDE
axis   ::= X | Y | DIAG_UL_LR | DIAG_UR_LL
```

Fig. 3. Spatial-program DSL. `Select` produces the initial scored candidate set $\mathcal{C}$ from the detector, and the remaining operators consume and produce candidate sets; operator roles and type signatures are given in Section IV-B, and the complete EBNF with production types in Appendix.

than the image plane. We retain this field algebra verbatim as one family of operators and add the missing set and order operators as a second family, so that the longer compositional expressions of Section III-E are synthesized from a common set of primitives rather than enumerated as new predicates.

B. Spatial-Program DSL

A program is a typed expression tree whose leaves generate candidates and whose internal nodes transform scored candidate sets. Fig. 3 gives the grammar. The leaf SELECT(t) queries the detector for noun phrase t and returns the initial scored set; FILTER reweights a set by a continuous geometric field; ARGMAX/ARGMIN, NTH, and RESTRICTCOUNT are the discrete set/order operators; RELATE scores a set by a binary spatial relation to an anchor; and AND/OR/NOT combine sets. SELECT has type $\mathsf{Phrase} \to \mathcal{C}$, the unary operators have type $\mathcal{C} \to \mathcal{C}$, and AND/OR have type $\mathcal{C} \times \mathcal{C} \to \mathcal{C}$, so the tree is well-formed exactly when its leaves are SELECT nodes and operator arities match. The operator set targets the expression space of remote-sensing referring, not a single benchmark: it covers position, extremum, ordinal, count, relation, and logical combination. Position and extrema dominate the two benchmarks studied here; the count, relation, and logical operators support the less frequent compositional expressions the domain admits. Table XIII maps a representative phrase to each.

1) Well-Formedness Checking: Before execution, a checker verifies that: 1) every candidate-producing argument ultimately roots in a SELECT leaf; 3) NTH carries a non-negative index and RESTRICTCOUNT a positive count; 3) the axis and polarity of an extremum are set; and 4) a binary relation that requires a second anchor (BETWEEN) supplies one. The checker returns the list of violated rules; a program with an empty list is *legal* and is executed; otherwise the ladder of Section IV-H takes over. A small deterministic repair pass runs first and fixes the most common synthesis slips without a model call: it drops a stray second anchor on a unary relation and promotes an anchor to the target when the nominal target is the image frame. Well-formedness checking and repair are model-free and cost microseconds.

2) Semantics: The program denotes a function from the image to a single box. Leaves are evaluated against the detector output; each internal node consumes the scored set(s) of its children and emits a scored set as defined below; and the referent is the top-scoring element of the root set, $b^\star = \arg\max_j s_j$. The type is closed under every operator, so fields and discrete operators interleave in any order, which is what allows ARGMAX(FILTER(SELECT (t), ϕ)) for a directional predicate ϕ, "the rightmost among those in the upper left," to be built from the same primitives as a bare directional filter.

C. Continuous Geometric Fields

The FILTER operator reweights a candidate set by a dense geometric field $P \in [0,1]^{H\times W}$ evaluated at each candidate. For positional predicates we reuse the closed-form, anchor-relative kernels of the field-only special case, which we summarize here for completeness; they are the *monotone* kernel family of Section IV-E. Let (c_x, c_y) be the center of anchor box A. For a cardinal direction d

$$q_d(x,y) = \begin{cases} x - c_x, & d = \text{right} \\ c_x - x, & d = \text{left} \\ y - c_y, & d = \text{bottom} \\ c_y - y, & d = \text{top} \end{cases}$$

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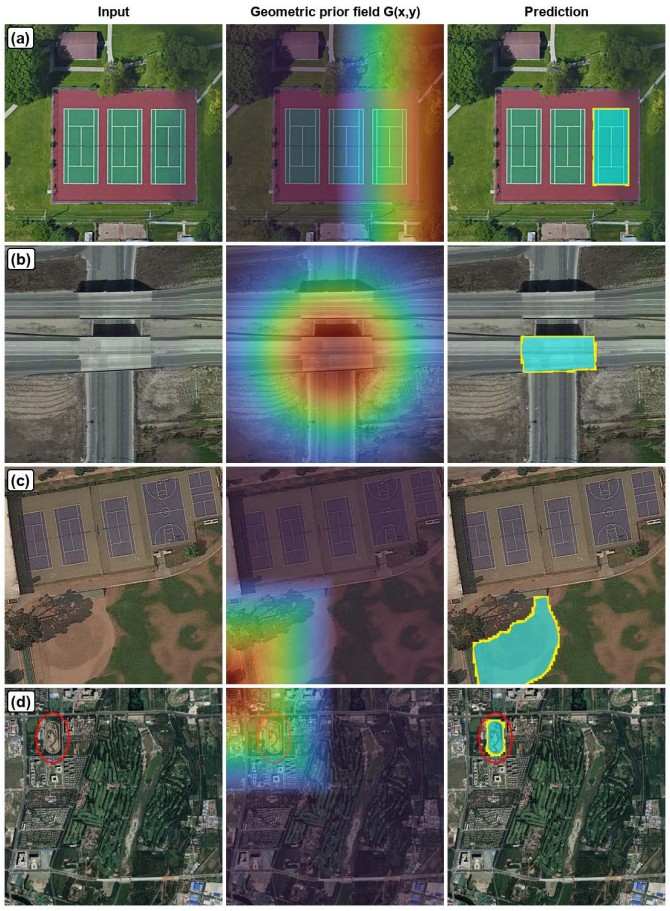


Fig. 4. Continuous geometric field $G(x, y)$ of the FILTER operator. For image-relative predicates, the field is computed in closed form against the image frame [(4) and (5)]; a candidate is scored at its center (6). Input (left), field as a heat map (middle, warm = high), and the resulting prediction (right; cyan = mask, yellow = ground truth). (a) Tennis court on the right. (b) Overpass in the middle. (c) Baseball field on the lower left. (d) Ground track field on the upper left.

$$P_d(x, y) = \left[\frac{\text{clip}(q_d(x, y), 0, L_d)}{L_d}\right]^{\gamma} \tag{4}$$

with $L_d = W$ for horizontal and $L_d = H$ for vertical directions and $\gamma = 1$ in the reported configuration. A diagonal field is the pixelwise minimum of its two cardinal fields. Scores are read at candidate box centers, which lie within the image domain $[0, W] \times [0, H]$, so the clip never has to separate two valid candidates; it bounds the normalization and does not change their ordering. The center and proximity kernels are

$$\begin{aligned} P_{\text{center}}(x, y) &= \exp\left[-\frac{(x-c_x)^2 + (y-c_y)^2}{2\sigma_c^2}\right] \\ P_{\text{near}}(x, y) &= \exp\left[-\frac{d_A(x,y)^2}{2L_A^2}\right] \end{aligned} \tag{5}$$

with $d_A(x, y)$ the Euclidean distance to the nearest point of A (zero inside the box), $\sigma_c = 0.25 \min(H, W)$, and $L_A = 1.5(w_A + h_A)/2$. A candidate box b_j with center (c_x^j, c_y^j) is scored by the field at its center

$$\rho(b_j) = P(c_x^j, c_y^j) \tag{6}$$

and FILTER updates its score multiplicatively, $s_j \leftarrow s_j \rho(b_j)$, so that a chain of filters realizes a conjunction of constraints. Center sampling makes the score independent of box size, which avoids the systematic dilution that a region mean inflicts on large referents (a big central object would otherwise score below a tiny box sitting on the prior peak). It also yields a useful invariance: because ρ reads a single point, the selection $\arg\max_j s_j$ is unchanged under any strictly increasing reshaping of a field. The directional exponent γ and the widths σ_c, L_A alter field magnitudes, but not the per-box ranking, so the headline is exactly invariant to them (Section V-H). These shape parameters thus drop out of the tuning surface; only the candidate set and the scoring rule remain as design choices. For an image-relative predicate, the anchor is the image frame; for an object-relative predicate, the anchor is resolved by recursively executing the anchor subprogram and taking its top-scoring box, which lets an anchor itself be constrained (*"right of the building in the upper left"*). AND, OR, and NOT combine scored sets matched by box identity

$$\begin{aligned} (\mathcal{C}_1 \wedge \mathcal{C}_2)_j &= \min\left(s_j^1, s_j^2\right) \\ (\mathcal{C}_1 \vee \mathcal{C}_2)_j &= \max\left(s_j^1, s_j^2\right) \\ (\neg \mathcal{C})_j &= 1 - s_j \end{aligned} \tag{7}$$

a noncompensatory conjunction in which a high response to one predicate cannot offset the violation of another; under center scoring, this min / max on scores equals the pixelwise min / max of the underlying fields, preserving the field-only equivalence. The complement $1 - s_j$ is composed under $\wedge$ with a positive constraint, so it acts as a soft exclusion mask that down-weights the excluded candidates and never stands alone as a selector. Fig. 4 visualizes the field and the grounding it produces for representative image-relative predicates.

Restricting the program to SELECT, FILTER, AND, and OR with the monotone kernel and the center score of (6) reproduces the field-only special case exactly, which fixes it as the reliability floor.

D. Discrete Compositional Operators

The set and order operators are the additions that cover the superlative, ordinal, and compositional expressions of the taxonomy, including a top-k restriction operator. Each is a closed-form map on the scored set; none introduces a learned parameter.

1) Extremum (Argmax / Argmin): A positional superlative selects the extreme candidate along an axis u, the signed projection of its center onto the a-direction (image x, image y, or an image diagonal). Hard selection is brittle under detector noise, so we instead *replace* the scores of the live set with a temperature-controlled softmax over u. The incoming score defines membership in the live set rather than acting as a weight: candidates an earlier filter drove to zero are excluded, and if none survive, the whole set is live. For j in the live set L

$$s_j \leftarrow \frac{\exp\left(\epsilon \beta u(b_j)\right)}{\sum_{k \in L} \exp\left(\epsilon \beta u(b_k)\right)}, \qquad \epsilon \in \{+1, -1\} \tag{8}$$

and $s_j = 0$ for $j \notin L$, with $\epsilon = +1$ for ARGMAX and -1 for ARGMIN. The temperature is set to $\beta = 1/\sigma_u$, the inverse standard deviation of the candidates' axis values $\{u(b_j)\}_{j \in L}$,

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so the softmax exponent is measured in spread units, and the operator is invariant to the absolute scale of the scene; a degenerate set ($\sigma_u = 0$, all candidates tied on the axis) gives uniform scores. Replacing rather than scaling is what stops a superlative from penalizing a candidate for its incoming score: the rightmost of a set of left-side ships is not docked for being rightmost. The result stays a graded set that a downstream operator can read. The axis polarity is fixed by the parser: *top-most* maps to ARGMIN on image y, and *right-most* to ARGMAX on image x. The reported configuration supports these positional superlatives, namely leftmost, rightmost, topmost, and bottommost. Metric-size superlatives and comparatives, such as *largest*, *smaller than*, and *similar in size*, are outside the claimed coverage. A noncomparative size adjective may remain in the detector-query noun phrase, but the system does not rank instances by size (see the Limitations in Section VI).

2) Ordinal (Nth): An ordinal expression (*"the second tank from the left"*) ranks the live candidates in reading order along the axis u and keeps the nth, with n a zero-based index supplied by the parser

$$s_j \leftarrow s_j \cdot \mathbb{1}\left[\operatorname{rank}_u\left(b_j\right) = n\right] \tag{9}$$

with rank_u ascending in $u(b_j)$, so the kept box retains its incoming score and the rest are zeroed; the operator prunes the set to one element while remaining type-compatible.

3) Top-k Candidate Restriction (RestrictCount): A count in the expression (*"the rightmost of the three ships"*) restricts the scored set to its top-k candidates by current score, with k from the parser

$$s_j \leftarrow s_j \cdot \mathbb{1}\left[\, j \in \text{top-}k\,(s)\,\right]. \tag{10}$$

RESTRICTCOUNT does not return a union of masks. It narrows the candidate set that a subsequent operator, typically ARGMAX/ARGMIN or NTH, selects from; when no count is given, it is the identity. The single-instance evaluation here scores one mask per expression; multimask output is controlled separately by the parser's group cardinality, which fires on a small number of group-tagged RISBench expressions.

4) Binary Relations (Relate): Beyond frame- and center-relative position, RELATE scores candidates by a binary spatial relation to an anchor set: NEAR, BETWEEN (two anchors), ADJACENT, CONTAINS, and INSIDE. Each relation is a closed-form score $r(b_j, A)$ on box geometry, *max-marginalized* over anchor instances and multiplied into the candidate score

$$\begin{aligned} s_j &\leftarrow s_j \cdot \max_{A \in \mathcal{A}} r\left(b_j, A\right) \\ r_{\text{inside}}\left(b, A\right) &= \frac{|b \cap A|}{|b|} \\ r_{\text{between}}\left(b, A_1, A_2\right) &= \exp\left[-\frac{d\left(c_b, \overline{c_{A_1} c_{A_2}}\right)^2}{2\sigma^2}\right] \end{aligned} \tag{11}$$

where c denotes a box center and $d(c_b, \overline{c_{A_1} c_{A_2}})$ is the perpendicular distance from the candidate center to the segment joining the two anchor centers (so BETWEEN peaks on that segment). The BETWEEN score is additionally gated by a betweenness factor that decays once the candidate projects past either endpoint of the segment, and its width scales with the geometry, $\sigma = 0.35|A_1A_2|$ plus the candidate's characteristic length, so the tolerance grows with anchor separation and target size. CONTAINS is $|b \cap A|/|A|$ and NEAR/ADJACENT use the proximity kernel of (5). Max-marginalization means a relation to "a building" is satisfied by the best-placed building.

E. Dual Geometric Kernels

The operators above draw their pixel- or axis-level scores from two kernel families with complementary shapes. The *monotone* kernel [(4) and (5)] increases without bound away from the anchor along the named direction and is correct for *furthest-in-direction* readings (*"the ship on the far right"*) and for the field-equivalent special case. The *extremal* kernel (8) is a softmax over the candidate set rather than over the image plane, and is correct for selecting one element by an axis extremum. Separating the kernels makes explicit that "left of" (monotone) and "leftmost" (extremal over the set) are different operations on the same axis, a distinction that a single directional ramp conflates and that we trace as a failure mode of the field-only special case (the "Failure Modes by Execution Route" analysis, Section V-K).

F. Executor

A legal program is evaluated by the recursive EXEC procedure of Algorithm 1. The executor walks the tree bottom-up: each SELECT leaf is bound to the detector's candidate set for its phrase; each operator node dispatches on its type to the corresponding map of Sections IV-C and IV-D; and AND/OR/NOT combine child sets by the field algebra. AND/OR/NOT match candidates by proposal identity, so their operands use the same normalized SELECT query, and its cached detections and the candidates align; reliable alignment across two different detector queries is outside the current scope. Anchor resolution follows one rule: a bare SELECT anchor is *marginalized* (its field is the disjunction over all its instances), whereas a *constrained* anchor subprogram is resolved to its single top-scoring box, so that "near a building" considers all buildings while "near the rightmost building" commits to one. The procedure is deterministic, allocates one $H \times W$ field per active filter, and runs in tens of milliseconds; its output is the ranked root set whose maximum is $b^\star$.

G. Program Synthesis

The synthesizer is a frozen, text-only instruction LLM (Qwen3-4B [48] by default) that reads E and emits the program as JSON. The model does not see the image, so the program is compiled from language alone, and perception enters only at the detector, executor, and segmenter. The prompt is a fixed, handcrafted few-shot schema, programmatically verified disjoint from the validation and test expressions; "thinking" decoding is disabled so the JSON is the first output. The schema encodes the design rules that a single pass otherwise gets wrong: positional superlatives map to the extremal operator with the correct axis polarity, image quadrants use the image frame as anchor, "middle" maps to CENTER (never to an extremum), and object-relative directional cues use the monotone kernel instead of a band. Synthesis is batched with

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left-padded greedy decoding, which is bit-identical to single-example decoding and amortizes the model load. Synthesis, checking, and execution are separated, so the parser can be swapped for a different LLM without touching the executor (Section V-F).

H. Reliability Ladder

Per-expression program synthesis trades the closed lexicon's guaranteed validity for expressive power, so we make the failure modes explicit and bounded. The ladder is: synthesize π; repair and check; if π is legal, execute it; if execution raises or returns an empty set, or if π is ill-formed, fall back to the field-only selector on the same candidates. The field-only selector is itself the monotone special case of Section IV-C, so its result is recovered exactly on every fallback; a well-typed program may still underperform it on an individual sample, but in aggregate the full system does not fall below it on either benchmark (Section V-E). We report the fraction of expressions resolved by an executed program with a nonempty result (the *program rate*) alongside accuracy, and measure the two separately because program success is not correctness (Section V). Every intermediate value (program, fields, scored sets, selected box) is inspectable, which makes the failure analysis of Section V-K possible.

I. Attribute-Conditioned Detection

Candidates come from an open-vocabulary detector (LAE-DINO [6]), queried for each SELECT phrase in attribute-conditioned form. The synthesis schema keeps color, shape, and noncomparative size adjectives inside the noun phrase (e.g., *"gray harbor"*) and routes supported positional cues into operators; a deterministic stabilizer then strips positional tails from detector queries and discards comparative-size tails, which are outside the reported scope. We run sliced inference with SAHI [49] at a 768-px tile size together with a full-image pass, use a detection threshold of 0.15, and apply per-class nonmaximum suppression; boxes are mapped to original image coordinates before execution. Detection is cached per image and query so that it is computed once and reused across program variants, since the candidate set depends only on the phrase and detector parameters and not on the rest of the program.

J. Segmentation

The selected box $b^\star$ prompts the Segment Anything ViT-L decoder [43] in single-mask mode, and the mask is scored under the protocol of Section III-F. The decoder is frozen and interchangeable; Section V-G shows the headline moves within a small band across SAM variants, confirming that the referring decision, not the segmenter, carries the result.

V. EXPERIMENTS

A. Setup

We evaluate on RRSIS-D [1], the standard RRSIS benchmark and single-instance by construction, and on RISBench [2], which is larger and has longer, more constrained expressions; both provide expression-conditioned pixel masks (Table I). MEET [50] contains 1.03 million zoom-free images for fine-grained geospatial scene classification, but its protocol provides scene labels rather than expression-conditioned masks and therefore does not support the mask-based evaluation here. RRSIS-D provides 17 402 image–expression pairs over 20 categories with a fixed train/val/test split; we use the 1740-expression val split for all design choices and the 3481-expression test split for the headline. RISBench is an order of magnitude larger, with 16 159 test expressions whose distribution is far richer in superlative, ordinal, and compositional constructions (Section III-E); it is used purely to measure transfer of the RRSIS-D-tuned configuration. This two-benchmark protocol follows recent work evaluated on both benchmarks [16], [37]. We report mIoU, oIoU, and Pr@X under the RMSIN protocol of Section III-F (eval@480 nearest).

The frozen configuration is a text-only Qwen3-4B synthesizer; LAE-DINO with SAHI tiling at 768 px and a detection threshold of 0.15; attribute-conditioned queries; the typed executor with the monotone and extremal kernels; and the Segment Anything ViT-L mask decoder in single-mask mode, with no learned reranker. All design choices, including the mask decoder and the few-shot schema, were made on RRSIS-D val; the configuration is frozen before any test

TABLE I

STATISTICS OF THE TWO EVALUATION BENCHMARKS. BOTH ARE SINGLE-INSTANCE REFERRING-SEGMENTATION BENCHMARKS WITH PIXEL MASKS; COUNTS ARE IMAGE–EXPRESSION–MASK TRIPLETS, ONE PER REFERRING EXPRESSION

Benchmark	Triplets	Categories	Image size	Split (train / val / test)		
				Train	Val	Test
RRSIS-D [1]	17,402	20	800×800	12,181	1,740	3,481
RISBench [2]	52,472	26	512×512	26,300	10,013	16,159

RRSIS-D expressions carry 7 attribute types; RISBench carries 8 and averages 14.31 words per expression (spatial resolution 0.1–30 m). All design choices are frozen on RRSIS-D val; the headline uses each benchmark's test split.

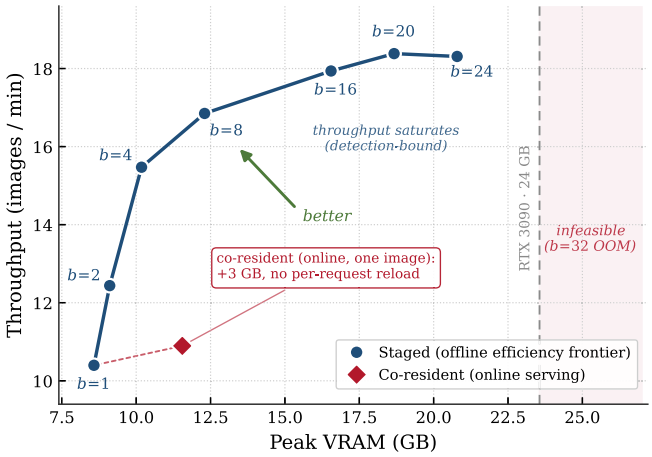


Fig. 5. Deployment efficiency on a single RTX 3090 (RRSIS-D val, $n = 200$, no caching): throughput versus peak VRAM across parsing batch sizes, with the co-resident point for one-image online serving.

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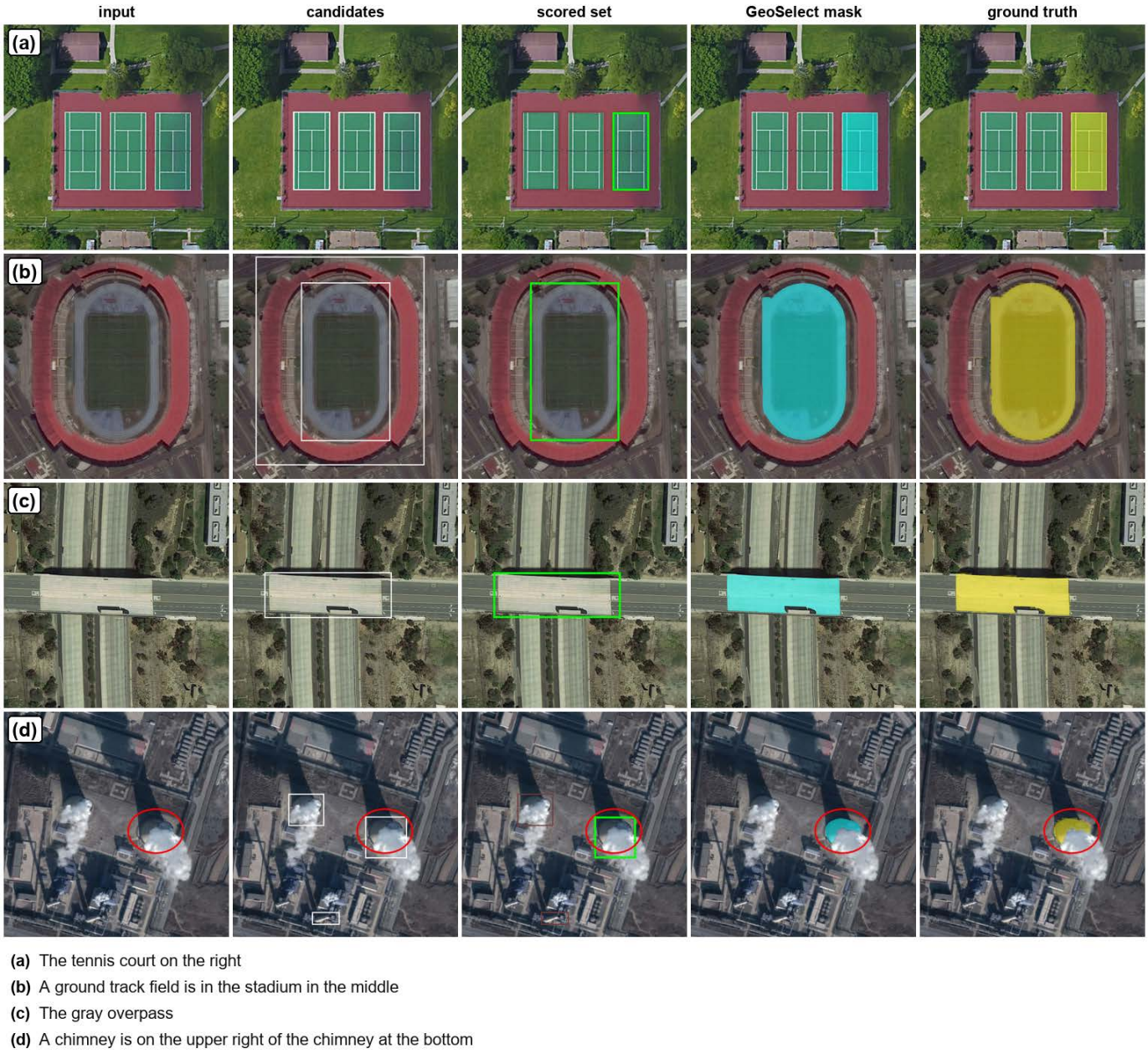


Fig. 6. Pipeline traces on RRSIS-D test, one case per row across the columns input, candidate boxes, scored candidate set (brighter = higher score, chosen box in green), GeoSelect mask (cyan), and ground truth (yellow). Rows span the strata: (a) image-relative, (b) and (d) object-anchored/compositional, and (c) attribute. Small targets are circled in red for visibility. The referring expression of each row is printed below.

evaluation, and no test-set recalibration is performed on either benchmark. The handcrafted few-shot prompt is programmatically verified to be disjoint from the val and test expressions. All inference runs on a single NVIDIA GeForce RTX 3090. Results for prior methods are taken from their source papers or published compilations [2], [14], [37]; we evaluate SAM3 ourselves.

GeoSelect achieves 58.86 mIoU / 59.48 oIoU on RRSIS-D test (Table II; mIoU 95% bootstrap CI $[57.71, 60.03]$): 2.08 times the best prior training-free method (RSVG-ZeroOV [16]) in mIoU and 2.6 times in oIoU, three times the decomposition-based EKP-HRM [15], and about 93% of the fully supervised specialist RMSIN [1] in mIoU. Its oIoU and mIoU are close, in contrast to the supervised specialists whose oIoU exceeds mIoU by 10–15 points; this reflects under-coverage of the largest referents by the single-mask decoder (Section V-G). Disabling the discrete operators (the field-only special case) gives 58.44, so on RRSIS-D, whose test split is almost entirely image-relative, object-anchored, and attribute expressions, the operators add little; their effect appears on RISBench (Section V-C) and on the object-anchored and compositional val strata (Table V). At stricter thresholds, GeoSelect reaches Pr@0.5/0.6/0.7/0.8/0.9 of 68.9/61.6/51.2/39.1/22.6 on the RRSIS-D test and 60.3/56.0/50.7/44.5/34.7 on the RISBench test. Fig. 6 shows qualitative predictions across strata, and Fig. 7 compares

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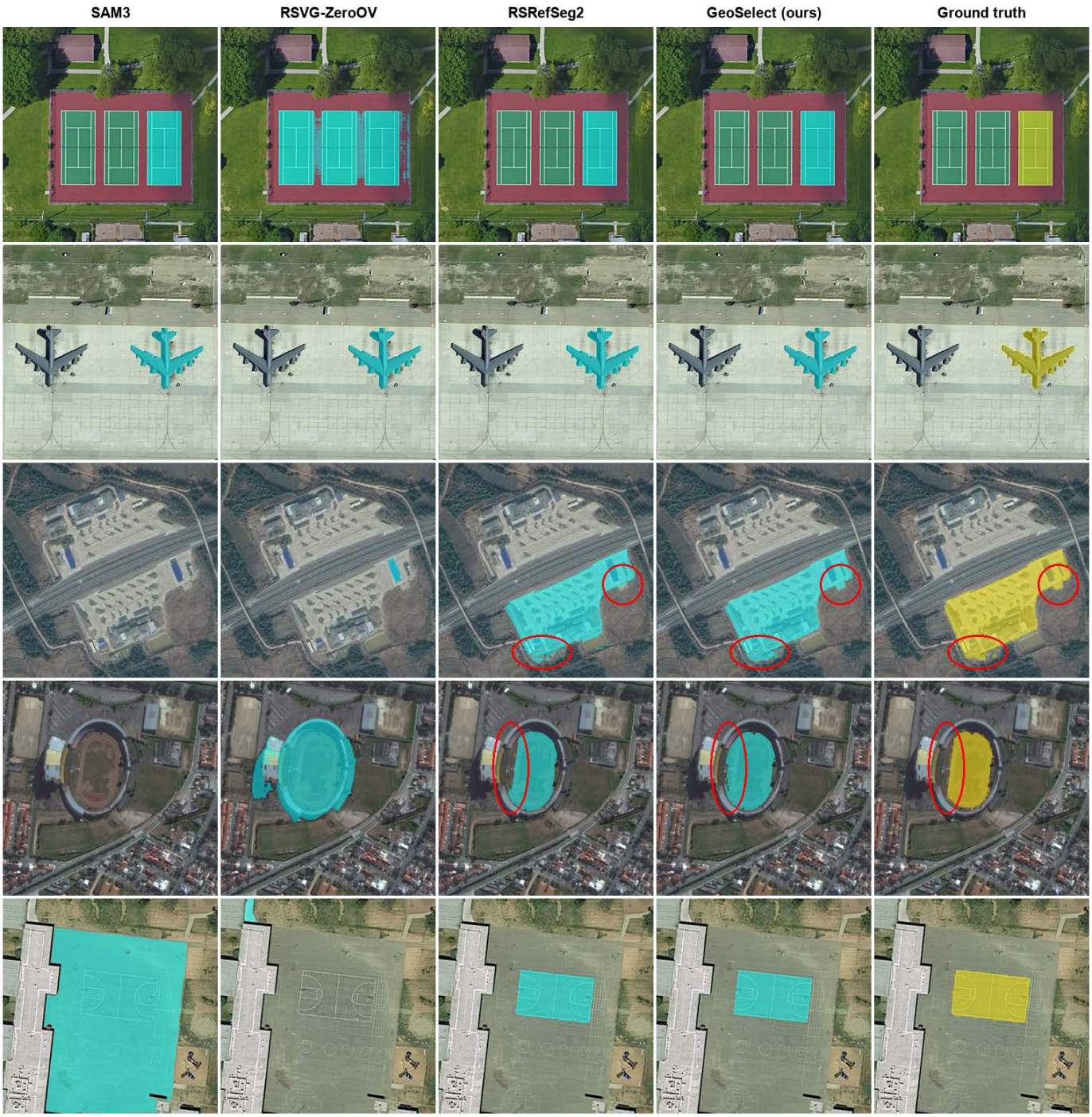


Fig. 7. Qualitative comparison on RRSIS-D test. Columns: SAM3 [18] (full-expression prompt), RSVG-ZeroOV [16], RSRefSeg2 [14] (supervised), GeoSelect (ours), and ground truth. Predictions in cyan, GT in yellow. The training-free baselines SAM3 and RSVG-ZeroOV over-segment multiple same-class regions or pick the wrong instance, whereas GeoSelect, also training-free, resolves the queried instance comparably to the supervised RSRefSeg2. Red ellipses mark the region that distinguishes the methods. The referring expression of each row is printed below the panel. (a) Tennis court on the right. (b) Airplane on the right. (c) Expressway service area is on the lower left of the overpass on the right. (d) Ground track field is in the gray stadium. (e) Rectangular basketball court.

GeoSelect with the training-free RSVG-ZeroOV and SAM3 and the supervised RSRefSeg2: the training-free baselines over-segment multiple same-class regions or select the wrong instance, whereas GeoSelect resolves the relation to the intended instance comparably to the supervised RSRefSeg2. GeoSelect updates no parameters on either benchmark (Table IV) and runs on a single NVIDIA GeForce RTX 3090. At batch size 1, the peak footprint is 8.6 GB, and the pipeline takes 5.8 s per image; batched parsing at batch size 16 (bit-identical predictions, so purely a throughput option) trades memory for speed, reaching 3.35 s per image at a 16.6-GB peak. Table III breaks the batched cost down by stage: open-vocabulary detection and LLM synthesis dominate (1.69 and 1.35 s), box-prompted SAM adds 0.29 s,

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TABLE II

MAIN RESULTS ON RRSIS-D TEST (n = 3481). FULLY SUPERVISED SPECIALISTS AND TRAINED VISION–LANGUAGE MODELS ARE AN UPPER REFERENCE, NOT TRAINING-FREE COMPETITORS; SOURCES AND EVALUATION PROTOCOL IN SECTION V-A. BOLD: BEST PER METHOD GROUP; (–) METRIC NOT REPORTED

Method	Venue	mIoU	oIoU
Fully-supervised specialists			
LAVT [8]	CVPR'22	61.12	76.48
LGCE [12]	TGRS'24	60.98	76.33
RMSIN [1]	CVPR'24	63.38	76.55
FIANet [3]	TGRS'24	64.01	76.81
CroBIM [2]	arXiv'25	64.24	76.37
ProVG [37]	arXiv'26	65.44	77.62
BTDNet [13]	arXiv'25	66.04	79.23
RSRefSeg2 [14]	TGRS'26	**69.17**	**79.45**
Trained vision–language models			
GeoGround [23]	arXiv'24	60.50	–
Text4Seg++ [24]	TPAMI'26	62.80	74.20
SegEarth-R1 [22]	arXiv'25	66.40	**78.01**
Sosa et al. (LoRA) [25]	arXiv'26	**67.6**	**–**
Training-free			
SAM3 (sentence prompt) [18]	arXiv'26	15.50	16.87
EKP-HRM [15]	GRSL'26	19.21	20.68
DGL-RSIS [17]	JAG'26	21.50	–
Sosa et al. (zero-shot) [25]	arXiv'26	24.9	–
RSVG-ZeroOV [16]	AAAI'26	28.35	22.83
GeoSelect (ours)	**–**	**58.86**	**59.48**

The field-only special case of our method and the discrete-operator gain are reported separately in the ablation of Table V (Section V-B). SAM3 is a concept segmenter; the full expression as prompt returns no instance for 69% of cases (15.50), and prompt formats are compared in Section V-G. Training-free numbers are quoted from each source paper. The two Sosa et al. [25] rows are its fully training-free (GPT-5 zero-shot) and LoRA-tuned (Qwen3-VL-2B) variants; only the former is training-free (Section II).

TABLE III

PER-STAGE INFERENCE LATENCY ON A SINGLE NVIDIA GEFORCE RTX 3090 (RRSIS-D VAL, n = 200, NO CACHING, PARSING BATCH SIZE 16). PROGRAM EXECUTION IS NEGLIGIBLE AGAINST THE FROZEN PERCEPTION MODULES

Stage	Time per image
LLM synthesis (Qwen3-4B)	1.35 s
Open-vocabulary detection (LAE-DINO)	1.69 s
Program execution (typed executor)	0.02 s
Segmentation (SAM ViT-L)	0.29 s
Total	3.35 s

and the typed executor costs only about 20 ms, so program execution is negligible against the frozen perception modules. Our reproduction of the diffusion-based RSVG-ZeroOV takes about 100 s per image on the same GPU. Fig. 5 plots the throughput–memory tradeoff across parsing batch sizes: it saturates once detection dominates, and the single-card operating range ends at batch size 24 before batch size 32 exceeds the 24-GB budget. The co-resident point is for one-image online serving: keeping the parser and segmenter both in memory avoids the per-request model reload that the staged batch-size-one path pays, so single-image throughput rises slightly at a cost of +3 GB.

TABLE IV

TRAINING COST. GEOSELECT UPDATES NO WEIGHTS AND CARRIES ONE FROZEN CONFIGURATION ACROSS DATASETS; THE SUPERVISED METHODS FINE-TUNE PER DATASET (RMSIN'S ∼200 M = SWIN-B + BERT-BASE)

Method	Paradigm	Trainable params	Per-dataset training
RMSIN [1]	supervised	full (∼200 M)	40 epochs
RSRefSeg2 [14]	supervised, LoRA	LoRA (PEFT)	300 epochs/dataset
RSVG-ZeroOV [16]	training-free	**0**	none
GeoSelect (ours)	training-free	**0**	none

RSRefSeg2 ships a separate checkpoint per benchmark; GeoSelect carries one. Inference latency is reported in the text.

TABLE V

SELECTOR COMPARISON ON RRSIS-D VAL (n = 1740): ALL VARIANTS USE THE SAME FROZEN DETECTOR AND SEGMENTER CONFIGURATION, AND DIFFER IN QUERY FORMULATION AND SELECTION MECHANISM. IM/OBJ/ATTR/COMP: IMAGE-RELATIVE/OBJECT-ANCHORED/PURE-ATTRIBUTE/COMPOSITIONAL. DETECTOR CONFIDENCE AND GEORSCLIP ARE IMPLICIT; RULE-BASED AND LLM-RANKING ARE SIMPLE EXPLICIT BASELINES; THE LAST TWO ARE OURS, WITH AND WITHOUT THE DISCRETE OPERATORS

Selector	mIoU	IM	OBJ	ATTR	COMP
Detector confidence	46.17	44.47	33.80	55.76	32.75
GeoRSCLIP [40]	45.22	41.90	35.23	56.50	34.57
Rule-based spatial	53.91	62.08	34.21	49.57	32.39
LLM candidate ranking	46.73	46.26	35.33	53.61	33.60
Ours, fields only	57.41	**63.34**	40.25	55.76	39.56
Ours, full program	**58.23**	63.24	**42.70**	**57.39**	**42.98**
Δ vs. best implicit	**+12.06**	+18.77	+7.47	+0.89	+8.41
Δ discrete vs. fields	+0.82	−0.10	+2.45	+1.63	+3.42

Explicit execution adds +12.06 mIoU over the best implicit selector (paired 95% CI [+10.37, +13.77]); the discrete operators add a further +0.82 (CI [+0.05, +1.58]). The rule-based and LLM-ranking baselines are simple explicit references that trail both variants on every stratum. Per-stratum operator effects on RISBench are in Table VI.

B. Explicit Execution Versus Implicit Matching

Under the same frozen detector and SAM ViT-L segmenter configuration, we vary the selector (RRSIS-D val, n = 1740; Table V). All methods use this same perception configuration, although their query formulation and candidate selection may differ, so these results compare complete grounding pipelines under a fixed perception stack. Explicit program execution improves mIoU by +12.06 over the stronger of the two implicit selectors (detector confidence at 46.17, above GeoRSCLIP region–text similarity at 45.22). Of this, the continuous field algebra alone accounts for +11.24 (field-only at 57.41), and the discrete operators add a further +0.82. The advantage lies on the spatial and compositional strata (+18.77 image-relative, +7.47 object-anchored, +8.41 compositional) and is near-zero on pure attribute (+0.89), where no spatial field exists, and the field-only selector reduces to the detector ordering. The ordering image > object follows anchor availability: the image frame is always present, whereas an object anchor must be detected and bound to the correct instance. The gain concentrates on the samples for which a spatial constraint can be executed, which is consistent

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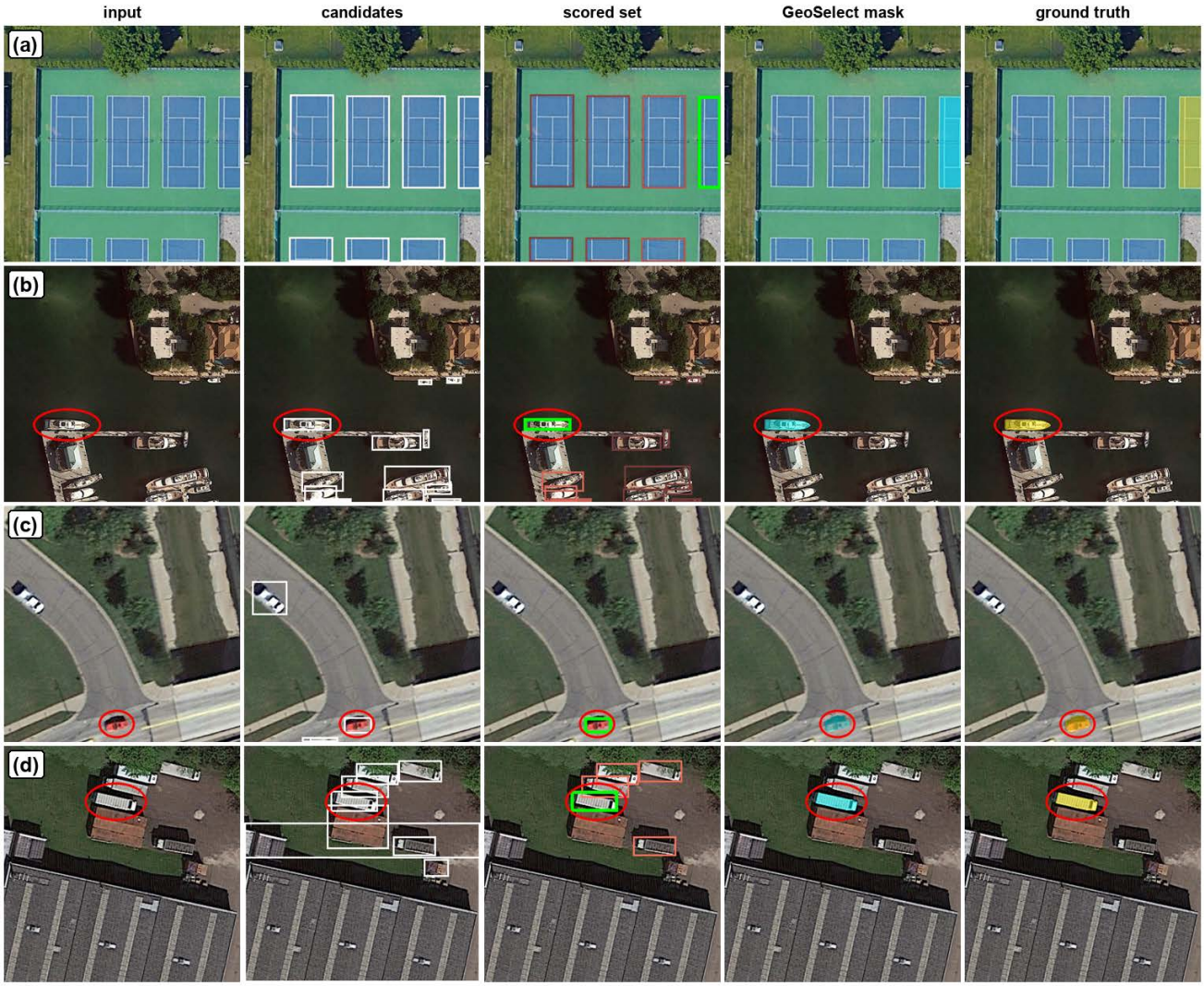


Fig. 8. Pipeline traces on RISBench test (frozen RRSIS-D configuration), one case per row across input, candidate boxes, scored candidate set (brighter = higher score, chosen box in green), GeoSelect mask (cyan), and ground truth (yellow). Rows span: (a) superlative, (b) compositional, (c) ordinal (*"the second vehicle"*), and (d) object-anchored proximity. Small targets are circled in red for visibility. The referring expression of each row is printed below.

with a benefit from explicit execution given the shared perception. It also scales with same-class clutter, the defining RRSIS difficulty: over that same stronger implicit selector, the advantage is only +1.7 mIoU when a single candidate is detected, rising to +12.4 at two or three candidates and +16.8 at four to six. Explicit execution helps precisely where multiple same-class instances compete. Two further selectors under the same configuration sharpen the comparison. The rule-based spatial selector applies one image-frame heuristic to the same candidates: a single directional cue takes the extreme box on that axis, a corner cue the diagonal extreme, "center"/"near" the box closest to the image center, and any relational, nested, or attribute-only expression falls back to the largest box. The LLM ranker uses the same Qwen3-4B and the same candidates, listed as normalized integer coordinates, and returns one candidate index directly, so the only difference from the full program is whether a typed program mediates the choice. They reach 53.91 and 46.73 mIoU, both below the field-only special case at 57.41 and the full program at 58.23. The rule-based selector is competitive where a single frame cue applies, reaching 62.08 on image-relative against 63.34 for field-only, and falls off to 34.21 on object-anchored and 32.39 on compositional expressions, which a fixed rule set cannot represent; neither hand-crafted spatial rules nor unconstrained LLM ranking matches typed execution.

C. Discrete Operators and Per-Stratum Gains

The bottom rows of Table V compare the full program with the field-only special case under the same configuration. On RRSIS-D val, they add +0.82 overall (paired 95% CI $[+0.05, +1.58]$), concentrated on object-anchored

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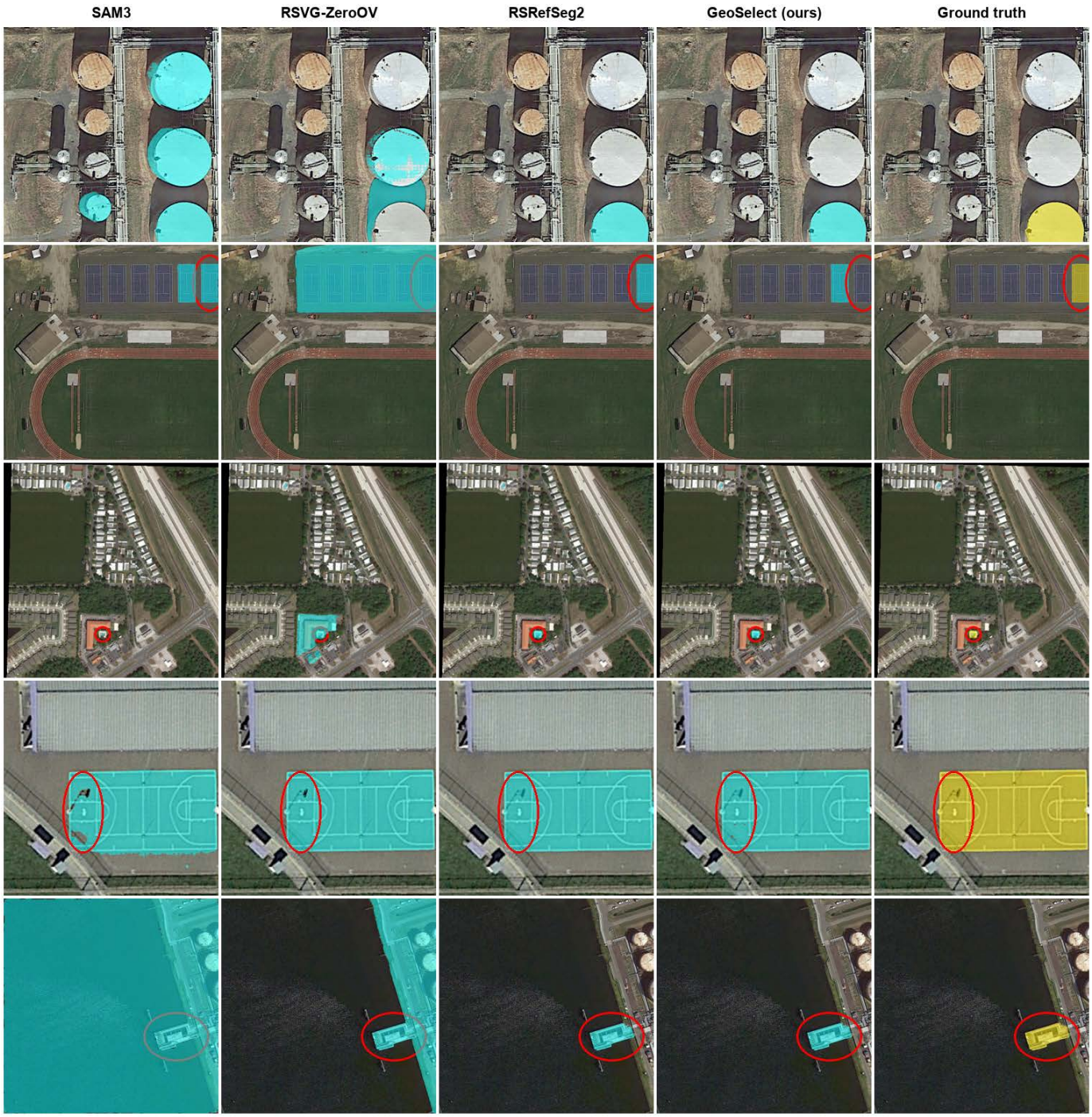


Fig. 9. Qualitative comparison on RISBench test. Columns: SAM3 [18] (full-expression prompt), RSVG-ZeroOV [16], RSRefSeg2 [14] (supervised), GeoSelect (ours), and ground truth. Predictions in cyan, GT in yellow. The longer, more constrained RISBench expressions widen the gap to the implicit baselines. Red ellipses mark a small or method-distinguishing target region. The referring expression of each row is printed below the panel. (a) Storage tank located at the bottom-right corner of the image. (b) Tennis court is at the top-right of the image. (c) Swimming pool adjacent to buildings with terracotta rooftops is located towards the bottom-middle of the image. (d) Basketball court is bordered by white lines and occupies a large rectangular area with its long side parallel to the nearby building. (e) Harbor extends into the water body with distinct linear features.

(+2.45) and compositional (+3.42) expressions and neutral on image-relative (−0.10), as expected since a pure directional field already handles the frame-relative case. Table VI resolves the gain by stratum on RISBench, where the operators are exercised at scale. The gain is +1.13 overall (95% CI $[+0.64, +1.63]$) and is carried by the superlative stratum (+2.22, $[+0.52, +3.84]$, $n = 2424$); on compositional, the operators match the field approximation to within noise (+0.63, $[-0.58, +1.86]$), and the ordinal stratum ($n = 52$) is too small to determine a sign. We therefore do not claim a uniform per-stratum improvement: the measurable accuracy gain concentrates on superlatives, while the operators' role on the remaining strata is one of coverage, since the field-only special case cannot represent these constructions at all

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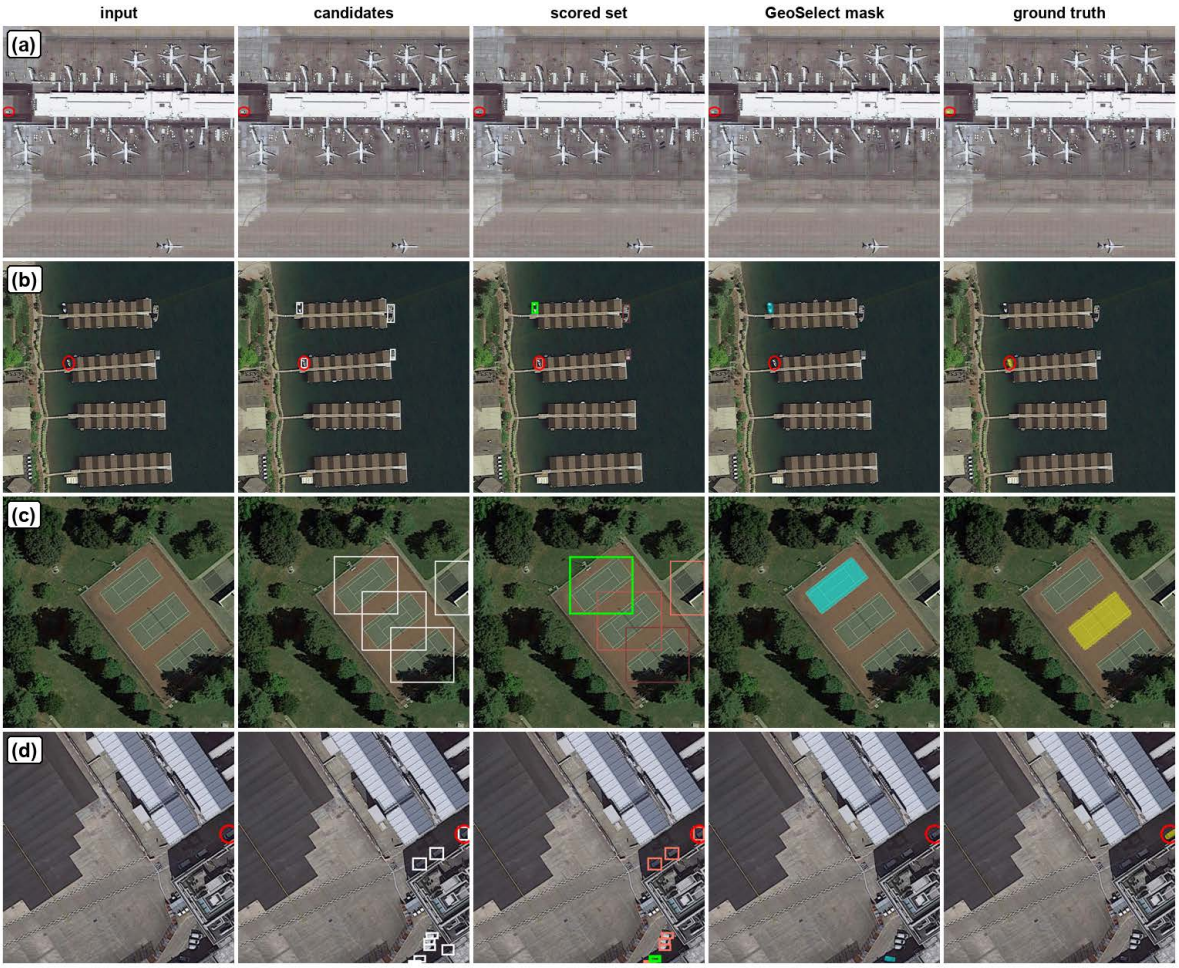


Fig. 10. Representative failure traces by execution route (cyan = prediction, yellow = ground truth; small targets circled in red). (a) Detection: The referent is too small to be proposed. (b) Selection: A near-perfect candidate exists, but the monotone ramp picks the wrong instance. (c) and (d) Program-path: The correct box is present, but the executed program selects the wrong instance. Per-case expressions and IoUs are in Section V-K.

(Section IV-D). The best-box-IoU analysis below reports how much of each stratum's available IoU the operators capture. Aggregate gains are modest because the field-only special case already resolves most frame-relative expressions, but on the constructions that require an operator the gap is large: across all 16 159 RISBench test expressions, 297 tagged as superlative or compositional have a field-only IoU below 0.1 and a full-program IoU above 0.9, for example *"the rightmost tennis court"* (0.00 → 1.00 via ARGMAX on x) and *"the bottom-most tennis court"* (0.00 → 0.99 via ARGMAX on y). The qualitative traces in Figs. 6 and 8 show one program-resolved case per stratum.

TABLE VI

DISCRETE-OPERATOR GAIN BY STRATUM ON RISBENCH TEST (n = 16 159): THE FIELD-ONLY SPECIAL CASE AGAINST THE FULL PROGRAM ON THE SAME CANDIDATES. Δ IS THE PAIRED MEAN IOU GAIN WITH A 95% BOOTSTRAP CI (SECTION V-C)

Stratum	n	fields	full	Δ (95% CI)
Superlative	2424	49.28	51.50	+2.22 [+0.52, +3.84]
Compositional	3998	49.17	49.81	+0.63 [−0.58, +1.86]
Ordinal	52	37.59	33.04	−4.55 [−17.3, +8.7]
Overall	16159	54.13	55.27	+1.13 [+0.64, +1.63]

Δ and its CI are computed from unrounded per-sample IoUs; the *fields* and *full* columns are rounded independently, so their printed difference can differ from Δ by 0.01.

D. Best-Box-IoU Headroom

The best-box-IoU proxy (Table VII) segments the best-IoU candidate in the detected pool. It reports how much IoU the proposed boxes already make available, a diagnostic for candidate availability; it does not measure program correctness. The candidate pool already contains a box above 0.5 IoU for 86.6% of RRSIS-D and 81.8% of RISBench expressions, so detection recall is high on both benchmarks. On RRSIS-D, GeoSelect captures 86.4% of this best-box IoU, and the best-box value itself reaches 68.16, just below the strongest supervised specialist RSRefSeg2 at 69.17 and above the others. Read only as a candidate-availability diagnostic, this motivates testing stronger in-domain proposals or finer tiling on RRSIS-D; it does not establish an error-free-ranking ceiling. The object-anchored stratum is the exception at 74.1% capture, a residual same-side selection loss of the kind in Fig. 10. On RISBench, the best-box IoU is much higher at 75.44, yet capture drops to 73.3%: the detected pool usually contains

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TABLE VII

PER-SAMPLE BEST-BOX-IOU PROXY: IOU FROM SEGMENTING THE BEST-IOU CANDIDATE IN THE DETECTED POOL. ACTUAL IS GEOSELECT; BEST THE PROXY; CAP. THE CAPTURED FRACTION. STRATA ABSENT FROM A BENCHMARK ARE OMITTED

Stratum	n	actual	best	cap.%
RRSIS-D test (n=3481)				
overall	3481	58.86	68.16	86.4
image	1853	64.05	70.08	91.4
object	628	43.18	58.29	74.1
attribute	1000	59.09	70.81	83.5
compositional	374	45.76	56.43	81.1
RISBench test (n=16159)				
overall	16159	55.27	75.44	73.3
image	11488	54.59	75.54	72.3
object	3949	55.62	75.70	73.5
attribute	722	64.11	72.34	88.6
superlative	2424	51.50	79.30	64.9
ordinal	52	33.04	71.99	45.9
compositional	3998	49.81	77.23	64.5

RRSIS-D test contains a negligible number of superlative/ordinal expressions ($n \leq 1$), so those strata are reported only on RISBench. Capture is computed from unrounded mIoU. The ordinal row (n=52) is small and reads with wide intervals (cf. the per-stratum CIs in Table VI); it is reported with its sample size and not over-interpreted.

TABLE VIII

SYNTHESIZER SCALING, CROSS-ARCHITECTURE, AND VISUAL INPUT ON RRSIS-D VAL (FROZEN CONFIGURATION; SWAP ONLY THE SYNTHESIZER). PROG% IS THE PROGRAM RATE: THE FRACTION OF EXPRESSIONS RESOLVED BY AN EXECUTED PROGRAM WITH A NONEMPTY RESULT

Synthesiser	Arch	mIoU	oIoU	prog%
Qwen3 family scaling (non-monotonic; 4B is the sweet spot)				
Qwen3-0.6B	dense Tf	49.57	51.72	94.4
Qwen3-1.7B	dense Tf	56.30	58.09	82.1
Qwen3-4B	dense Tf	**58.23**	**59.30**	82.4
Qwen3-8B	dense Tf	54.28	54.70	78.4
Cross-architecture (∼3–7B)				
Qwen3-4B	dense Tf	**58.23**	**59.30**	82.4
Phi-4-mini [51]	dense Tf	57.05	57.93	78.8
Falcon-Mamba-7B [52]	SSM	55.12	57.18	81.7
Llama-3.2-3B [53]	dense Tf	54.90	55.84	45.1
Visual input (text-only vs. vision-language, 4B class)				
Qwen3-4B	dense Tf	**58.23**	**59.30**	82.4
Qwen3-VL-4B [54] (text)	VLM	56.73	57.38	80.0
Qwen3-VL-4B (+image)	VLM	56.73	58.04	81.1

Giving Qwen3-VL-4B the image leaves mIoU unchanged (56.73), moves oIoU by +0.66, and moves the program rate by +1.1 points. The text-only Qwen3-4B stays ahead of both vision-language variants on mIoU.

a usable box, but a larger share of it goes uncaptured than on RRSIS-D, and that uncaptured share concentrates on the new strata. Pure attribute captures 88.6%, while superlative at 64.9%, compositional at 64.5%, and ordinal at 45.9% on n = 52 leave the most room. Since a usable box is usually present on these strata, the natural place to look for further training-free gains on RISBench is the program path: axis estimation, anchor binding, and ranking on compositional expressions. The proxy reports candidate availability and does not decompose the error causally, so we read this as an investigation priority (Section VI).

E. Reliability

Per-expression synthesis is made safe by the reliability ladder of Section IV-H. The program rate is the fraction of expressions resolved by an executed program with a nonempty result, with the rest falling back. It is 83.4% on RRSIS-D test, or 2904 of 3481, and 91.0% on RISBench test, or 14 698 of 16 159. The lower RRSIS-D figure comes from legal programs whose candidate set is empty: the open-vocabulary detector returns no proposal for the queried phrase. The remaining expressions fall back to the field-only selector and recover its result exactly, and in aggregate the full system stays above this field-only special case by +0.42 on RRSIS-D test and +1.13 on RISBench (Table VI). The program rate measures successful program resolution, that is, synthesis, checking, execution, and a nonempty result, rather than correctness, and the two are reported separately: a 0.6B parser reaches the highest program rate yet the lowest accuracy (Section V-F). A blind manual check of 100 legal-program outputs on RRSIS-D val, stratified across operator types to cover the rare strata, finds 98% of the synthesized programs to be correct interpretations, the two exceptions being size comparatives outside the spatial scope (Section VI). The same check on 100 RISBench val outputs finds 92% correct. The eight errors are two size superlatives resolved by position alone, two dropped positional cues, one wrong target noun, one wrong direction, one positional phrase used as an object anchor, and one spurious nesting level. These two numbers are not directly comparable to the program rate: the manual audit measures semantic correctness conditional on a stratified sample of legal-program outputs, whereas the program rate measures how often synthesis, checking, and execution complete with a nonempty result over the full split. In these stratified samples of legal-program outputs, semantic correctness was 98% on RRSIS-D val and 92% on RISBench val.

F. Parser Capacity and Architecture

The synthesizer is a lightweight, swappable component (Table VIII). Within the Qwen3 family, scaling is nonmonotonic and peaks at 4B: the 0.6B parser loses 8.7 mIoU despite the highest program rate (94.4%), and the 8B parser (54.28) sits below the 4B, so the task rewards reliable schema following over raw capacity. Across architectures, a dense transformer (Phi-4-mini, 57.05) and a state-space model (Falcon-Mamba, 55.12) both stay within about 3 mIoU of the Qwen3-4B headline, and Llama-3.2-3B reaches 54.90 even with a low 45.1% program rate because the reliability ladder routes its many ill-formed programs to the field-only fallback. The method therefore requires a parser of at least about 2B that follows the DSL schema reliably, but not a specific architecture: dense transformers and a state-space model are interchangeable at the headline. A vision-language synthesizer does not raise the headline either (bottom block of Table VIII): Qwen3-VL-4B [54] scores 56.73 mIoU with and without the image, below the text-only Qwen3-4B.

G. Mask Decoder

The box-to-mask stage is replaceable. We use SAM ViT-L in single-mask mode, size-aligned (308 M) with the alternatives. Swapping only this stage on RRSIS-D val, with synthesis,

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TABLE IX

PARAMETER ANALYSIS ON RRSIS-D VAL (n = 1740), ONE PARAMETER VARIED AROUND THE FROZEN VALUE (BOLD). FIELD-SHAPE PARAMETERS (γ, σ_c, L_A) ARE OMITTED: CENTER SCORING (EQ. 6) MAKES SELECTION INVARIANT TO THEM, SO EVERY VALUE YIELDS 58.23 EXACTLY

Parameter	value	mIoU	oIoU
Scoring rule			
`score_method`	region-max	57.90	59.22
	region-mean	58.07	58.73
	centre	**58.23**	**59.30**
Detection threshold			
τ_{det}	0.10	57.20	57.90
	0.15	**58.23**	**59.30**
	0.20	58.60	59.90
	0.25	57.80	59.84
SAHI tile size			
tile	512	46.76	41.52
	768	**58.23**	**59.30**
	1024	58.27	59.83

detection, and selection frozen and every decoder in single-mask mode, gives SAM ViT-L 58.23, SAM 2.1 hiera-large 56.26, and SAM3 with a box prompt 53.22. The decoder moves the headline within a 5.0-point band, far smaller than the +12.1 that explicit execution adds over implicit matching (Section V-B), so the result is not tied to a particular segmenter. SAM ViT-L edges SAM 2.1 because, on large referents, SAM 2.1's single mask tends to return a part instead of the whole object; SAM3, designed for short concept prompts, does not surpass it. SAM ViT-L is selected on val and used for the reported test result.

H. Parameter Analysis

Center scoring (6) collapses the tuning surface. Each candidate is scored at a single point, making the selection invariant to any strictly increasing reshaping of a field, so the directional exponent γ, the center width σ_c, and the proximity length L_A change no prediction. Sweeping each over a wide range ($\gamma \in [0.5, 2.0]$, $\sigma_c \in [0.15, 0.35]$) leaves the headline at 58.23 mIoU to the reported precision, an exact confirmation of the analytic invariance of Section IV-C rather than an insensitivity that tuning could exploit. The extremal temperature β is likewise inert on this split, which has essentially no superlative expressions for it to act on. The parameters that remain (Table IX) are the scoring rule and the candidate set.

The scoring rule confirms the center-sampling design: center (58.23), edges region-mean (58.07), and region-max (57.90). The gain concentrates on the object-anchored and compositional strata, where large referents make the region mean dilute the prior. Pure-attribute is identical across all three rules because a uniform field reduces every rule to the detector ordering. The detection threshold defines a flat plateau (57.2/58.2/58.6/57.8 at 0.10/0.15/0.20/0.25) on which the frozen 0.15 sits within 0.4 mIoU of the val-optimal 0.20; we keep the preregistered value rather than reselect the val maximum, a sub-CI difference that would not justify rerunning the test and transfer suite. The SAHI tile shows the only sharp dependence: 768 (58.23) and 1024 (58.27) are equivalent, but 512 collapses to 46.76 because tiles below the referent scale starve detection of recall, confirming the frozen 768 is well chosen and that the candidate set, not the field algebra, is the sensitive component.

TABLE X

IMAGE-ID-LEVEL SEEN/UNSEEN AUDIT ON RRSIS-D TEST (DETECTOR–BENCHMARK OVERLAP; MEMBERSHIP IS DETECTOR-LEVEL AND PIPELINE-INDEPENDENT, SO THE SEEN-SETS APPLY UNCHANGED AND THE METRICS REPORTED HERE ARE GEOSELECT'S OWN). UNSEEN DENOTES SAMPLES OUTSIDE THE SPECIFIED SEEN-SET; THE UNSEEN MIOU IS THE EXPOSURE-AUDITED NUMBER

Seen-set	seen%	SEEN		UNSEEN	
		mIoU	oIoU	mIoU	oIoU
Exact (gold)	32.8	59.71	58.60	**58.44**	59.91
Conservative	54.3	62.56	61.47	**54.46**	56.35

Two seen-set definitions: the exact gold set (images the detector actually trained on) and a conservative worst case (every RRSIS-D test image in DIOR trainval marked seen). Analysis in Section V-I.

I. Pretraining-Exposure Audit

The open-vocabulary detector is pretrained on LAE-1M [6], which overlaps RRSIS-D's source imagery (both draw on DIOR, with LAE-1M including DIOR trainval while holding out its test split). We quantify this exposure at the image-id level under two seen-set definitions (Table X); the membership is detector-level and pipeline-independent, so the audit rebuckets GeoSelect's own per-sample IoUs. On the exact gold seen-set, the seen−unseen gap is +1.27 mIoU and the unseen subset (58.44) is close to the full result. We read this descriptively: on this image-id audit, the difference is small, though it does not rule out other forms of pretraining exposure. The conservative seen-set is a worst case (it marks every RRSIS-D test image in DIOR trainval as seen regardless of which LAE-1M actually sampled), and its unseen subset still reaches 54.46. Both unseen subsets remain well above prior training-free full-test results. A per-category breakdown of the same audit is given in Table XI: unseen meets or exceeds seen in 9 of the 20 mapped categories. An in-domain detector remains important for recall: LAE-DINO reports that generic open-vocabulary detectors with natural-scene pretraining reach only about 1 AP_{50} on DIOR, against 82–86 once pretrained on LAE-1M. Within GeoSelect, the detector supplies category- and attribute-level proposals; the referring decision is made by the executed program.

We audit RRSIS-D only. RISBench is built from DIOR and DOTA-v2 [2]; the same DIOR and DOTA imagery is among LAE-1M's largest training sources [6], so its detector exposure is plausibly at least as high as RRSIS-D's, and we make no exposure-robustness claim for RISBench. An image-id-level audit is moreover not feasible on the released RISBench: its imagery is re-encoded as renamed 512 × 512 crops without back-references to the source DIOR/DOTA-v2 IDs, and the correspondence between RISBench and source-dataset splits is undocumented, so neither ID matching nor content hashing against the originals is reliable. This exposure question

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TABLE XI

PER-CATEGORY MIOU ON RRSIS-D TEST BY PRETRAINING EXPOSURE (3418 CATEGORY-MAPPED EXPRESSIONS, 63 PROGRAM-NULL EXCLUDED; SEEN/UNSEEN SPLIT FROM TABLE X). BOLD: THE 9 OF 20 CATEGORIES WHERE UNSEEN MEETS OR EXCEEDS SEEN. THE WEIGHTED UNSEEN MIOU HERE IS 58.64; THE ALL-EXPRESSION 58.44 IS IN TABLE X

Category	UNSEEN		SEEN	
	mIoU	n	mIoU	n
Airplane	**75.97**	54	66.90	45
Airport	53.65	160	57.57	36
Baseball field	63.72	135	75.30	94
Basketball court	**73.06**	76	71.05	44
Bridge	**50.75**	195	43.50	67
Chimney	74.22	75	77.58	32
Dam	**61.18**	133	53.75	42
Expressway service area	69.29	67	76.00	51
Expressway toll station	64.93	93	72.73	21
Golf field	54.37	87	55.96	24
Ground track field	53.61	168	55.82	66
Harbor	25.70	16	36.54	11
Overpass	**55.18**	158	48.38	63
Ship	66.73	115	69.68	62
Stadium	**75.93**	63	74.19	25
Storage tank	**71.47**	68	64.04	39
Tennis court	66.78	45	77.27	57
Train station	**64.16**	105	61.81	23
Vehicle	45.10	306	47.83	238
Windmill	**55.44**	185	55.34	74

TABLE XII

CROSS-DATASET TRANSFER TO RISBENCH TEST (n = 16 159) UNDER THE FROZEN RRSIS-D CONFIGURATION, WITH NO TEST-SET RECALIBRATION. SUPERVISED ROWS: RISBENCH LEADERBOARD COMPILED IN RSREFSEG2 [14]/CROBIM [2]. BOLD: BEST PER METHOD GROUP

Method	Venue	mIoU	oIoU
Fully-supervised specialists (RISBench leaderboard [2])			
RRN [9]	CVPR'18	43.18	49.67
BRINet [10]	CVPR'20	42.91	48.73
CMPC+ [11]	TPAMI'21	46.73	53.98
CRIS [7]	CVPR'22	55.18	69.11
LAVT [8]	CVPR'22	61.93	74.15
RMSIN [1]	CVPR'24	63.07	74.09
CARIS [55]	MM'23	65.79	**75.10**
CroBIM [2]	arXiv'25	67.32	73.61
RSRefSeg2 [14]	TGRS'26	**72.57**	74.77
Training-free			
SAM3 (sentence prompt) [18]	arXiv'26	21.09	26.20
RSVG-ZeroOV [16]	AAAI'26	31.84	26.35
GeoSelect (ours)	–	**55.27**	**55.88**

The field-only special case of our method and the per-stratum discrete-operator gain (+1.13 mIoU overall) are reported in the ablation of Table VI (Section V-C). SAM3 returns no instance for 64% of cases here.

is, however, orthogonal to the attribution of our gain: the comparison of Section V-B uses the same frozen detector and segmenter for every selector, so its +12.06 mIoU advantage is not explained by detector quality or by any pretraining exposure, both shared across selectors. Pairing the executor with the strongest available aerial-pretrained open-vocabulary detector is a deliberate part of the training-free design. The audit bears only on the absolute headline, where the RRSIS-D gold-set gap of +1.27 mIoU is small on this measure.

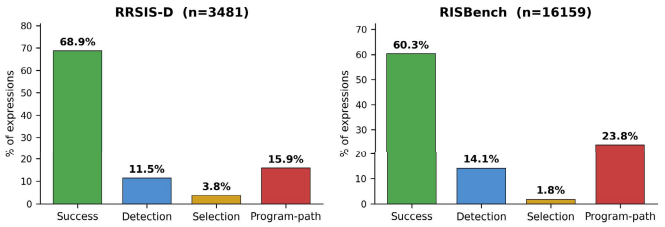


Fig. 11. Outcome partition by execution route on the two test sets, using the *Success*/*Detection*/*Selection*/*Program-path* outcomes defined in Section V-K. Each names the route that produced the reported mask, not a causal stage diagnosis; their cross-dataset shift is read in Section VI.

J. Cross-Dataset Transfer: RISBench

The full configuration is selected before test evaluation and applied to RISBench with no test-set recalibration, so Table XII measures transfer. GeoSelect reaches 55.27 mIoU/55.88 oIoU on RISBench test (mIoU 95% bootstrap CI [54.64, 55.89]), 1.74 times the best prior training-free method in mIoU and 2.1 times in oIoU, at about 88% of RMSIN. With no RISBench supervision, it surpasses the early LSTM-based supervised methods RRN, BRINet, and CMPC+ on both metrics. The gain decomposes into the in-domain candidate pipeline and a further +1.13 from the discrete operators, larger than on RRSIS-D because RISBench is rich in the constructions they target (Section V-C). The two-dataset scope matches recent practice: RSVG-ZeroOV and the concurrent ProVG [37] evaluate on exactly these two. Figs. 8 and 9 show qualitative predictions across strata and the five-way baseline comparison; the gap to the implicit baselines widens on RISBench's longer, more constrained expressions.

K. Failure Modes by Execution Route

Every intermediate is inspectable, so we can trace each outcome to an execution path (Fig. 10). We partition every expression by the route that produced its final mask: *Success* covers IoU at least 0.5; *Detection* covers expressions where no candidate reaches IoU 0.5; *Selection* covers those where the field-only fallback is used, yet the result stays below 0.5; and *Program-path* covers those where an executed program is used, yet the result stays below 0.5. Fig. 11 reports the share of each outcome. *Detection* (a): the referent is too small to be proposed, so the candidate set is empty and no prior, program, or mask can recover it (best − box = 0); this is an unrecoverable proposal-recall failure, and higher resolution tiling or a stronger detector directly targets this route. *Selection* (b): a near-perfect candidate exists (best-box IoU 0.85) but several instances satisfy "on the left" and the monotone ramp scores the furthest-left one rather than the annotated instance. A graded directional field cannot encode whether "on the left" denotes a region or an extremum (the monotone-versus-extremal distinction of Section IV-E), so this ambiguity is a limitation of the field representation rather than a tunable error. *Program-path* (c) and (d): the correct box is present (best-box IoU 0.94/0.66) but the executed program selects the wrong instance. In (c), the program is well-formed and keeps the nested anchor for "above the court on the lower right," yet the resolution binds to the wrong tennis court; in (d), a multicue

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phrase collapses to a single predicate. Such program-path errors, which span synthesis, execution, and ranking, are the most frequent unsuccessful outcome on RISBench (Fig. 11), where the detected pool usually already contains the right box. The mask decoder contributes comparatively little: swapping it moves the headline within a 5.0-point band (Section V-G).

VI. Discussion and Limitations

A. Where the Headroom Is

The best-box-IoU proxy (Section V-D) segments the best-IoU candidate in the detected pool. It is a candidate-availability diagnostic: it reports how much IoU the proposals already make available, and it is neither a per-sample upper bound nor a decomposition into synthesis, execution, ranking, and mask decoding. Read as a directional signal, it gives a benchmark-specific picture of where the remaining error sits. On RRSIS-D, the proxy leaves a 9.30-point gap between the actual mIoU of 58.86 and the best-box value of 68.16, and that best-box value itself lies just below the strongest supervised specialist; the outcome partition in Fig. 11 assigns 11.5% of RRSIS-D expressions to detection and 15.9% to the program path. Proposal recall and program-path behavior are therefore complementary targets on RRSIS-D: a missed proposal is unrecoverable, while the program path accounts for the larger share of unsuccessful outcomes. On RISBench, the balance shifts toward the program path. Detection transfers well (the best-box IoU rises to 75.4), but the proxy gap widens to 20.17 points, the program-path share rises to 23.8%, and the uncaptured share concentrates on the superlative, ordinal, and compositional strata, where the correct box is usually already among the candidates. This makes the program path the clearer priority on RISBench (Section V-D). The discrete operators move these strata well above the field-only floor, and the 25–40-point gap to the best-box IoU on them marks the most promising direction for further training-free gains: better axis estimation under detector noise, anchor binding for nested expressions, and a quantitative spatial-consistency check on the synthesized program. Because the proxy is noncausal, we read these as investigation priorities rather than isolated error sources.

B. Is the Program Machinery Necessary?

The field-only special case already doubles the best prior training-free result, which invites the question of whether the DSL and its discrete operators earn their place. The answer is yes, at two levels. The accuracy the field-only case reaches is itself a product of the same executed representation. Under the same frozen detector and segmenter, executing a geometric program beats the stronger of the two implicit selectors by +12.06 mIoU (95% CI $[+10.37, +13.77]$, Section V-B), consistent with explicit execution over the field algebra contributing the bulk of the gain. On top of that, the discrete operators extend the same calculus to constructions the field algebra structurally cannot express, and deliver a resolved +2.22 mIoU on RISBench superlatives (Table VI). The field-only number is thus not evidence against the apparatus but a lower rung of it.

C. Expressiveness–Reliability Tradeoff

Replacing a closed predicate lexicon with per-expression program synthesis buys the compositional coverage of Section V-C at the cost of programs that can be ill-formed or semantically wrong. The well-formedness checker and the reliability ladder bound this downside (Section V-E). This is the structural difference from prompt-era visual-programming pipelines that dispatch to opaque neural subroutines: our symbolic half is an explicit geometric calculus whose every intermediate is inspectable and whose failures are catchable.

D. Methodological Scope

The few-shot schema was developed on RRSIS-D val and includes grammar rules (extremal axis polarity, image-quadrant anchoring, "middle" mapped to CENTER) that are general language-to-program mappings rather than dataset-specific tuning, and the configuration transfers to RISBench with no test-set recalibration. We report it as designed on RRSIS-D val and validated on RISBench rather than as design-free, since some compositional constructions are rare on RRSIS-D.

E. Limitations

GeoSelect targets the single-instance regime, exposed through the parser's cardinality field; generalized $0/1/k$ referring [56] is out of scope. Absolute metric thresholds, such as *"within 50 m,"* require a known ground-sample distance and are also out of scope. Size and comparative-size expressions such as *"smaller than"* are outside the claimed coverage: size is orthogonal to the directional field algebra, and the only size signal available downstream of the text-only synthesizer is the detected bounding-box area, an unreliable proxy for semantic size. A noncomparative size adjective may remain in the detector-query noun phrase, but the system does not rank instances by size. It assumes an in-domain open-vocabulary detector, without which proposal recall collapses (Section V-I). The axis-aligned box geometry underlying the extremal axes is orientation-sensitive for elongated objects such as ships and bridges: on RISBench superlatives, IoU falls from 57.9 for near-square selected boxes (aspect ratio ≤ 1.5) to 23.9 for elongated ones (ratio > 3, $n = 298$). Soft ranking mitigates but does not remove this; an oriented-box geometry is a clear next step. Finally, the ordinal stratum is small on current benchmarks ($n = 52$ on RISBench), so its large best-box-IoU gap, while suggestive, rests on too few samples to interpret.

VII. Conclusion

We reframed training-free RRSIS as explicit spatial-program execution. A frozen, text-only language model synthesizes a typed program in a small DSL, a well-formedness checker accepts it, and a deterministic executor runs it over a single scored-candidate-set type that unifies continuous geometric fields with discrete set and order operators. A reliability ladder degrades any failing program to the field-only special case, so the full system stays at or above the field-only floor in aggregate. GeoSelect reaches

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58.86 mIoU on RRSIS-D and 55.27 mIoU on RISBench, more than twice and about 1.7 times the prior training-free results. Under a fixed detector and segmenter, explicit execution improves RRSIS-D val by +12.1 mIoU over implicit selectors. The discrete operators make the compositional constructions expressible that the field algebra cannot and add a resolved gain on superlatives; the remaining headroom splits between proposal recall and the program path, more of it on the program path for RISBench. The method targets single-instance expressions and assumes an in-domain detector. Extending the field algebra to at-a-distance and metric-relative readings, such as a tuned banded kernel for "next to" and scale-relative distance thresholds, is left to future work. Executing an explicit spatial program brings training-free RRSIS to about 93% of the supervised specialist RMSIN on RRSIS-D while keeping every intermediate decision inspectable.

APPENDIX
SYNTHESIS SCHEMA AND REPRODUCIBILITY

The synthesizer uses a single fixed few-shot prompt, released verbatim with the code. It declares the operator grammar of Fig. 3, the image-coordinate polarity (X grows left-to-right and Y top-to-bottom, so *topmost* maps to ARGMIN on Y and *bottommost* to ARGMAX on Y), and a fixed set of language-to-program rules: image-frame phrases ("on the right," "in the lower left") map to a FILTER anchored to the image; "middle"/"center" maps to the CENTER predicate and never to an extremum; corner phrases map to a single diagonal predicate ("lower left" → LL); positional superlatives map to ARGMAX/ARGMIN on the axis the superlative word selects; "*n*th from" maps to NTH; an object-relative direction ("left of the harbor") maps to a monotone FILTER with a nested anchor, while RELATE is reserved for NEAR/BETWEEN/ADJACENT/CONTAINS/INSIDE; and color, shape, and noncomparative size adjectives stay inside the SELECT noun phrase rather than becoming predicates. The prompt is verified disjoint from the validation and test expressions at the string level (exact match after lowercasing and whitespace normalization). It was developed on RRSIS-D val and applied unchanged to RISBench. The prompt is deliberately over-specified for safety: keeping only the prose rules (no examples), only the examples (no rules), or only four of the fourteen examples each leaves *every* prediction byte-identical on RRSIS-D val, so the reported accuracy does not depend on prompt micro-tuning. The prompt is scaffolding, not a contribution; the substantive ablation of the executed representation is the fields-only-versus-full comparison (Section V-C, Table VI), not the prompt wording. Table XIII lists the explicit phrase-to-operator mapping.

Grammar (EBNF): The DSL is the typed grammar below. SELECT maps a detector-query noun phrase to the initial scored candidate set $\mathcal{C}$; the unary operators map $\mathcal{C} \to \mathcal{C}$ and AND/OR map $\mathcal{C} \times \mathcal{C} \to \mathcal{C}$, so the root returns a scored candidate set whose top-scoring box is passed to the segmenter.

```
program ::= expr
expr ::= Select(phrase)
| Filter(expr, pred, anchor)
| Argmax(expr, axis) | Argmin(expr, axis)
| Nth(expr, index, axis)
| RestrictCount(expr, count)
| Relate(expr, rel, anchor [, anchor])
| And(expr, expr) | Or(expr, expr) |
Not(expr)
anchor ::= IMAGE | expr
pred ::= LEFT | RIGHT | TOP | BOTTOM
| UL | UR | LL | LR | CENTER | NEAR
rel        ::= NEAR | BETWEEN | ADJACENT |
CONTAINS | INSIDE
axis ::= X | Y | DIAG_UL_LR | DIAG_UR_LL
index              ::= non-negative integer
(zero-based)
count ::= positive integer
phrase ::= detector-query noun phrase
```

Here, `phrase` retains color, noncomparative size, and shape adjectives and is passed verbatim to the detector; `Nth` takes a zero-based index and `RestrictCount` a positive count, and `Relate` takes one anchor, or two for BETWEEN. Fig. 3 summarizes the same grammar graphically.

Worked Execution Examples: Two examples trace a program from expression to the selected box, where DETECT returns the candidate set for each SELECT phrase.

1) *"the rightmost tennis court"* synthesizes to ARGMAX(SELECT("tennis court"), X). SELECT returns the detected tennis courts as a scored set; ARGMAX on the

TABLE XIII

REPRESENTATIVE PHRASE-TO-OPERATOR MAPPINGS FROM THE FIXED SYNTHESIS PROMPT (IMAGE POLARITY: X LEFT-TO-RIGHT, Y TOP-TO-BOTTOM). THE LEAF SELECT FOR THE HEAD NOUN PHRASE IS ELIDED WHERE AN OPERATOR APPLIES TO IT, AND THE OPERANDS OF AND / OR SHARE ONE SELECT QUERY. SEE APPENDIX FOR THE FULL EBNF

Expression phrase	Operator skeleton (head SELECT elided)
"a dam on the right"	FILTER(RIGHT, IMAGE)
"in the middle / central area"	FILTER(CENTER, IMAGE)
"in the lower left"	FILTER(LL, IMAGE)
"the rightmost airplane"	ARGMAX(X)
"the leftmost airplane"	ARGMIN(X)
"the bottommost tank"	ARGMAX(Y)
"the topmost tank"	ARGMIN(Y)
"the second ship from the left"	NTH(n=1, X)
"the ship to the left of the harbor"	FILTER(LEFT, SELECT(harbor))
"a vehicle above the toll station"	FILTER(TOP, SELECT(toll station))
"the rightmost of the three tanks on the left"	ARGMAX(RESTRICTCOUNT(FILTER(LEFT, IMAGE), k=3), X)
"the building between the river and the road"	RELATE(BETWEEN, SELECT(river), SELECT(road))
"the car near the building"	RELATE(NEAR, SELECT(building))
"a light gray windmill"	SELECT("light gray windmill")
"a stadium near the river and in the upper left"	AND(RELATE(NEAR, SELECT(river)), FILTER(UL))
"a ship on the left or near the top"	OR(FILTER(LEFT), FILTER(TOP))
"a storage tank not near the building"	AND(SELECT("storage tank"), NOT(RELATE(NEAR, SELECT(building))))

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x-axis keeps the single rightmost box and zeroes the others, reducing the set to one element; that box is segmented.

2) *"the ship on the left near the top"* synthesizes to

$$\text{And}\big(\text{Filter}(\text{Select}(\text{“ship”}), \text{Left}, \text{Image}) \\ \text{Filter}(\text{Select}(\text{“ship”}), \text{Top}, \text{Image})\big).$$

Both operands query the same "ship" detections, so their candidates carry identical box identities; each FILTER multiplies the shared scores by its directional field, AND takes the per-candidate minimum, and the ship that is both on the left and near the top is selected. In each case, the output is a single-instance mask.

Reproducibility: The code, the frozen configuration, the complete prompt, the per-sample IoU dumps, and the bootstrap-CI script used for all confidence intervals are publicly available at https://github.com/Avalon-S/GeoSelect, so the reported numbers can be recomputed without rerunning detection or segmentation.

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